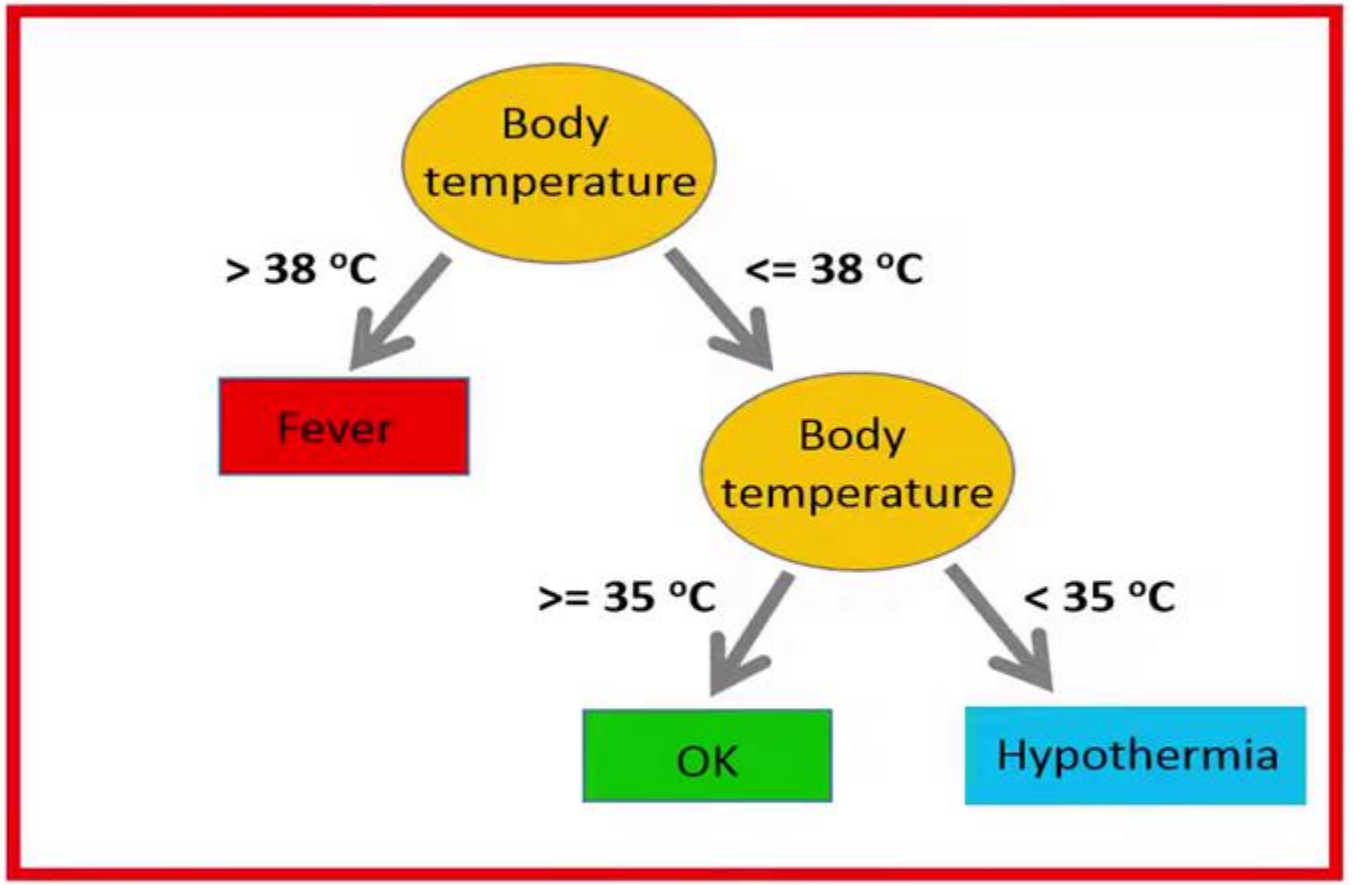
Random Forests

Dr.Hafeez Anwar

Random forest



Random forest is a method for classification, or regression, which is based on decision trees.

Decision tree

Cancer

Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

Let's say that we have the following data of nine men, where four of these are known to have prostate cancer, and five are known to be healthy.

Decision tree





	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

For each person, we have information about the age,

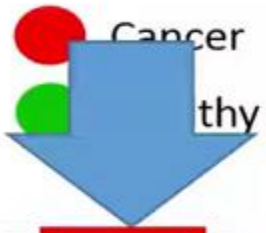
Decision tree



	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

the prostate-specific antigen concentration in the blood,

Decision tree



	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

as well as a score from one to five based on an MRI scan of the prostate, where a score of one looks normal whereas a score of five is a strong indication of cancer.

Decision tree

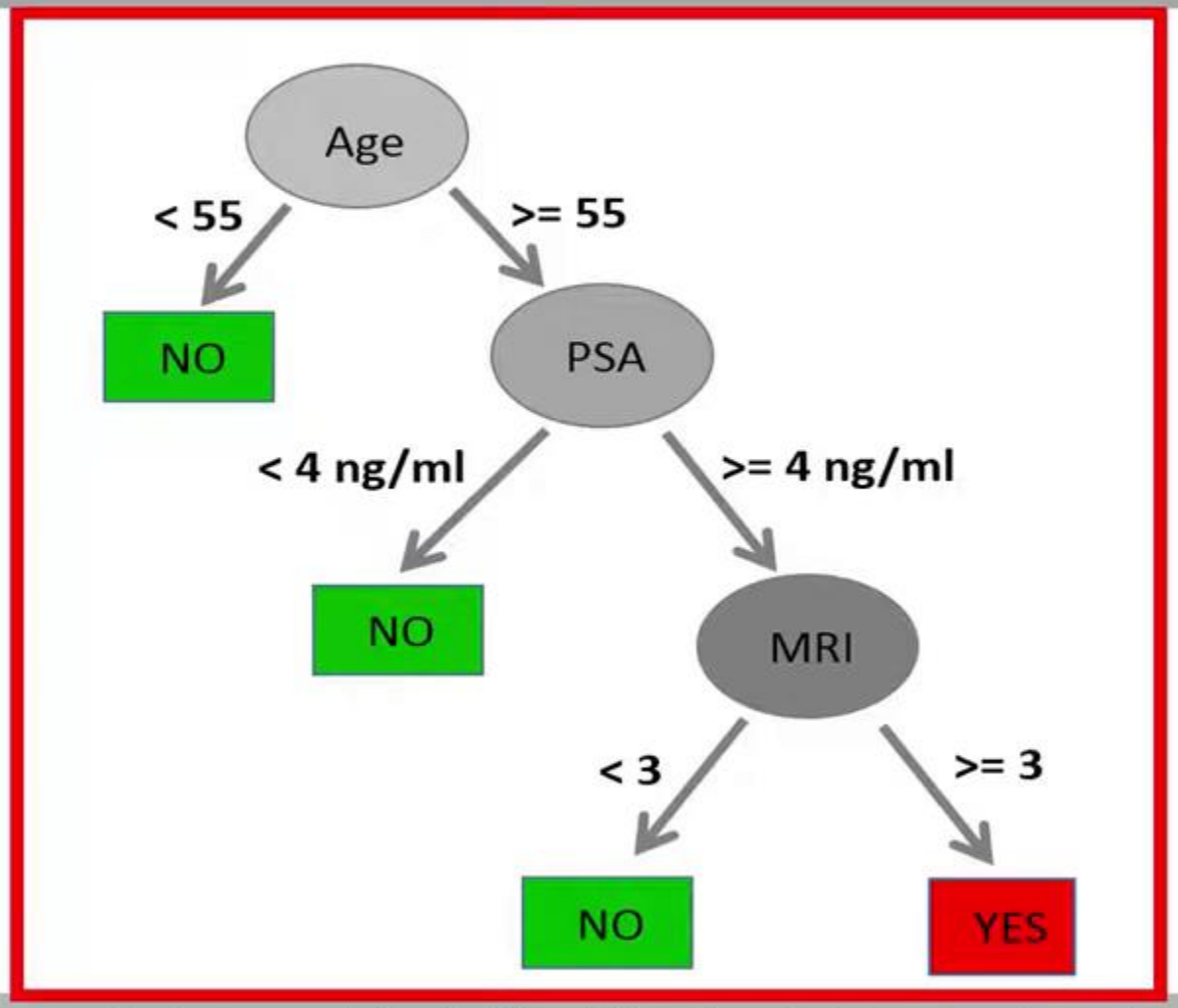
Cancer

Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

Note that we need a lot more data points to create a useful decision tree and to run a random forest. I have here only used nine individuals because it will then be easier to explain how the method works.

Decision tree

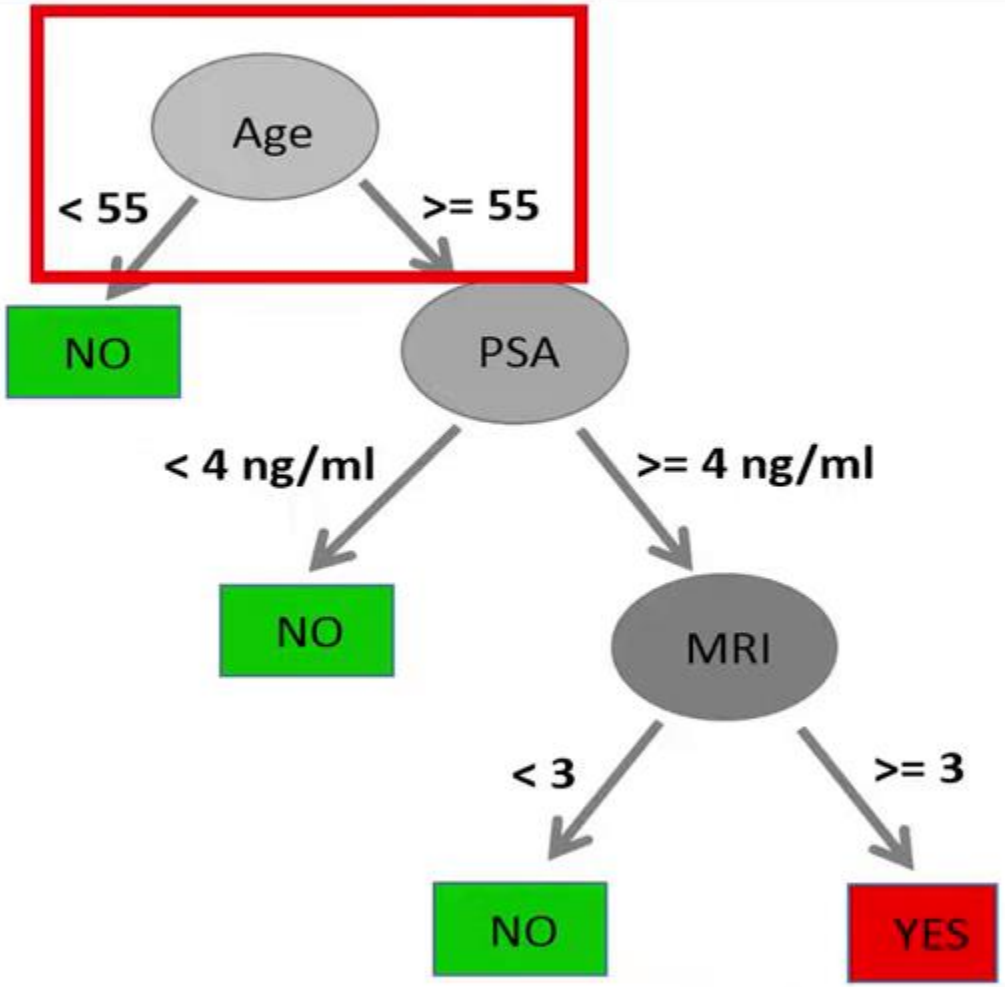


Cancer
Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

Suppose that we would build the following decision tree based on all data points. This tree is used to predict if someone has prostate cancer or not.

Decision tree

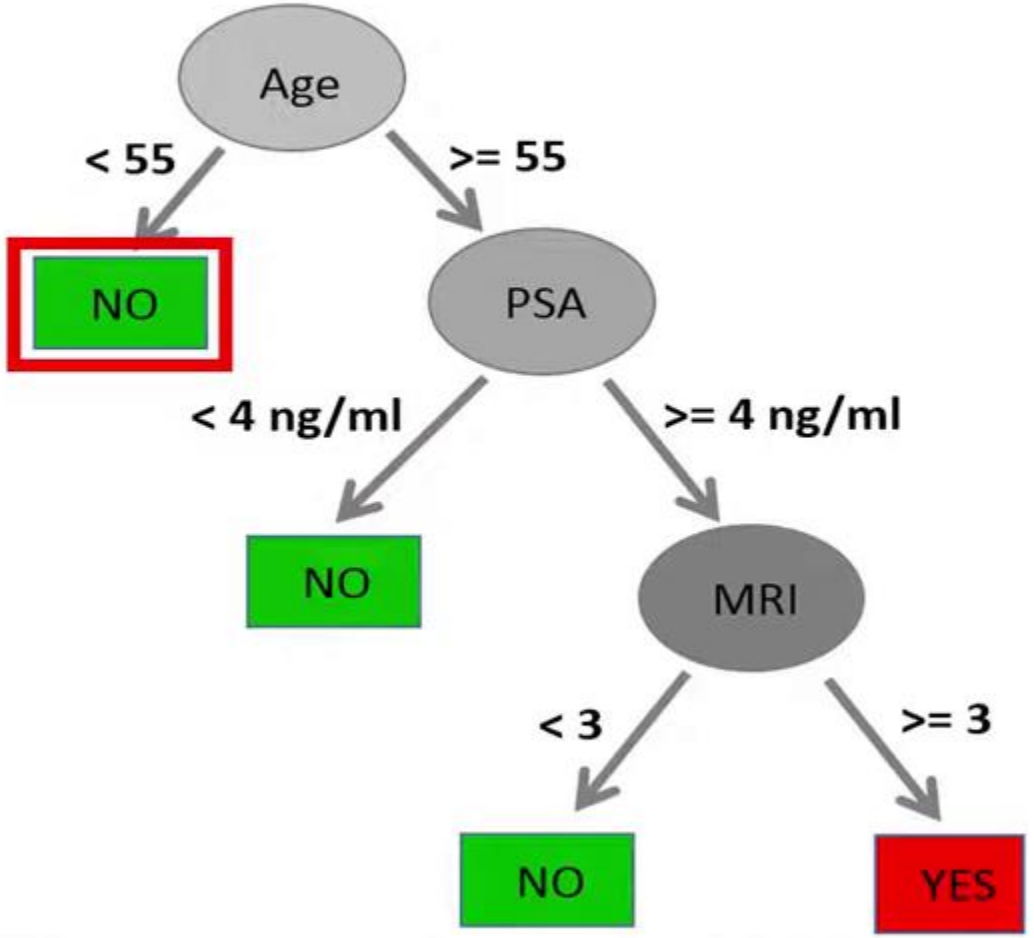


Cancer
Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

If a male is younger than 55 years of age, it is quite unlikely that the person has prostate cancer.

Decision tree

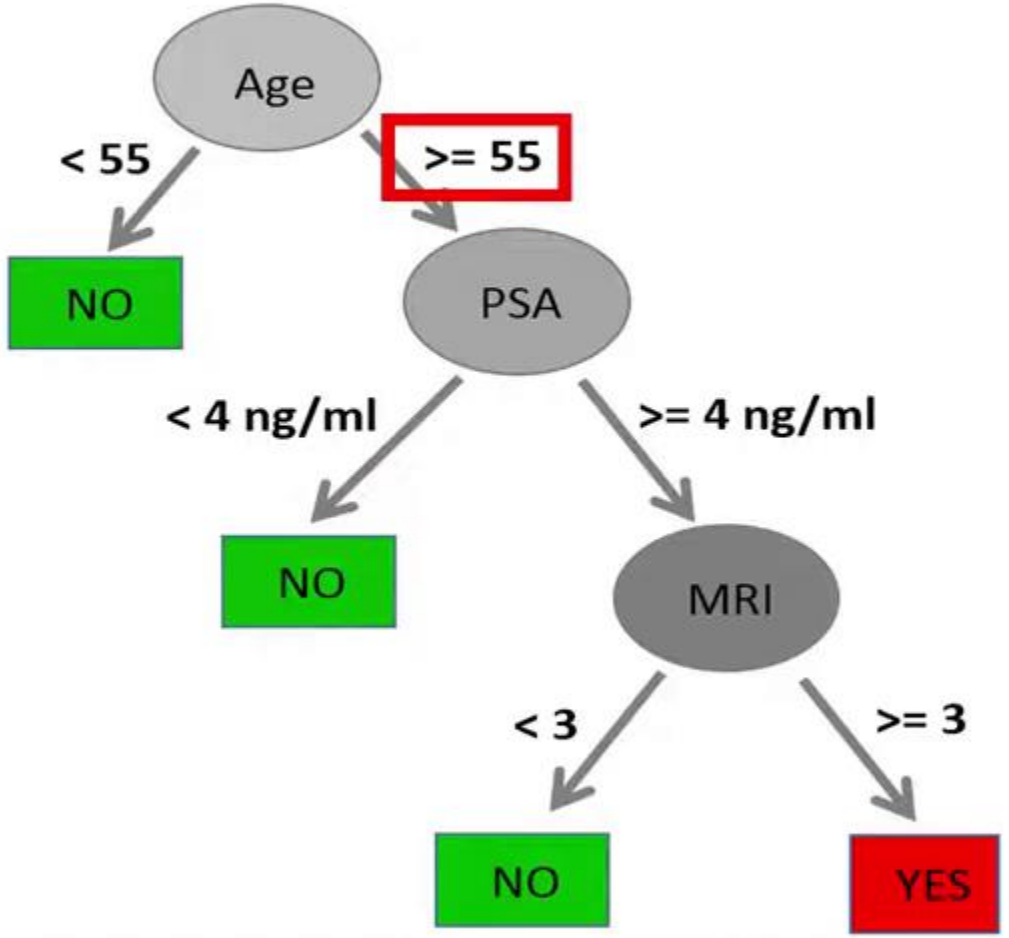


Cancer
Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

The tree therefore predicts that the person does not have prostate cancer.

Decision tree

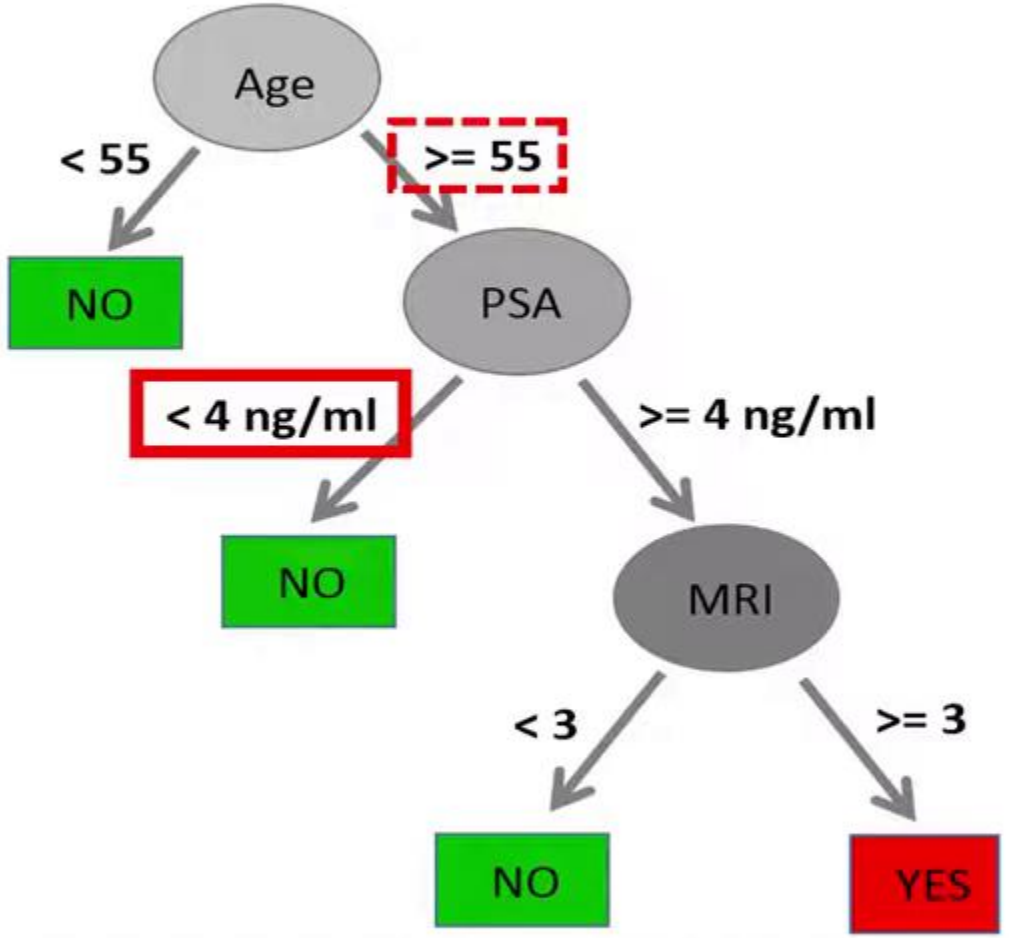


Cancer
Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

If the person is 55 years, or older,

Decision tree

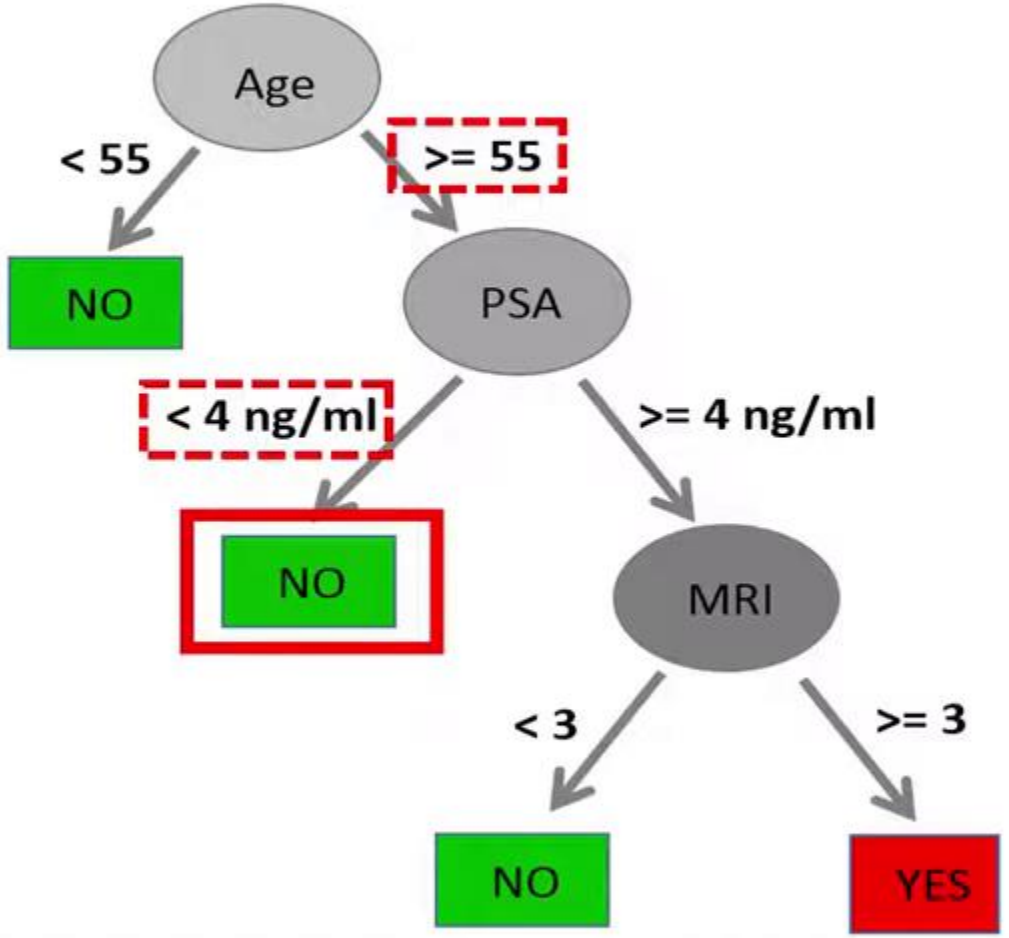


Cancer
Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

but has a PSA level that is less than 4 ng/ml,

Decision tree

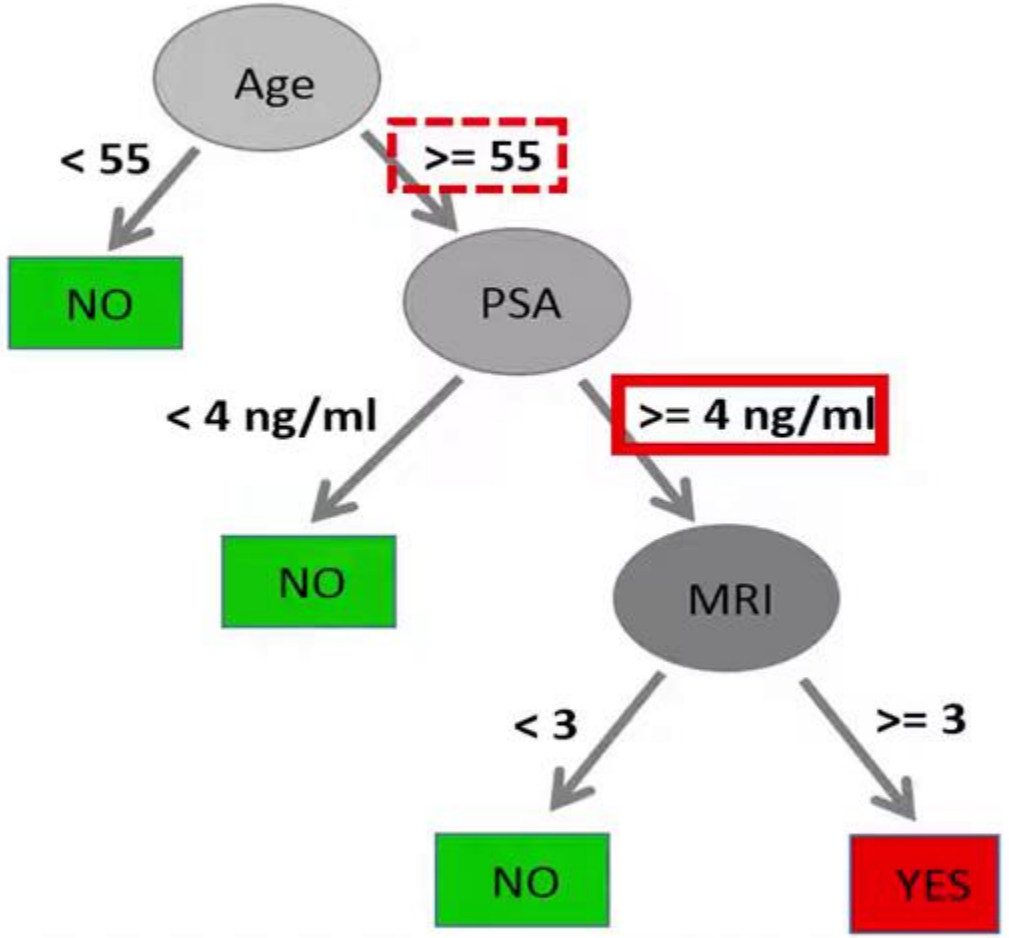


Cancer
Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

the person is predicted to not have prostate cancer.

Decision tree

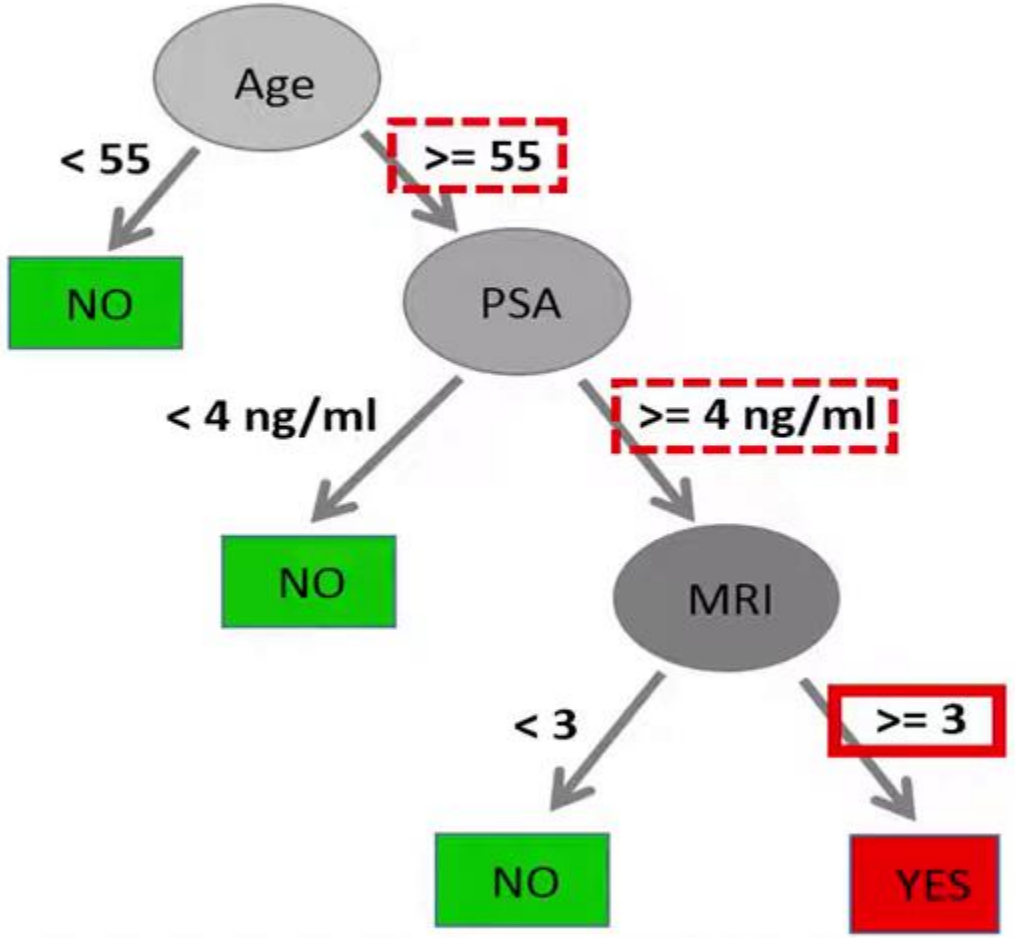


Cancer
Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

However, if the person has a PSA level that is equal to, or greater than 4,

Decision tree

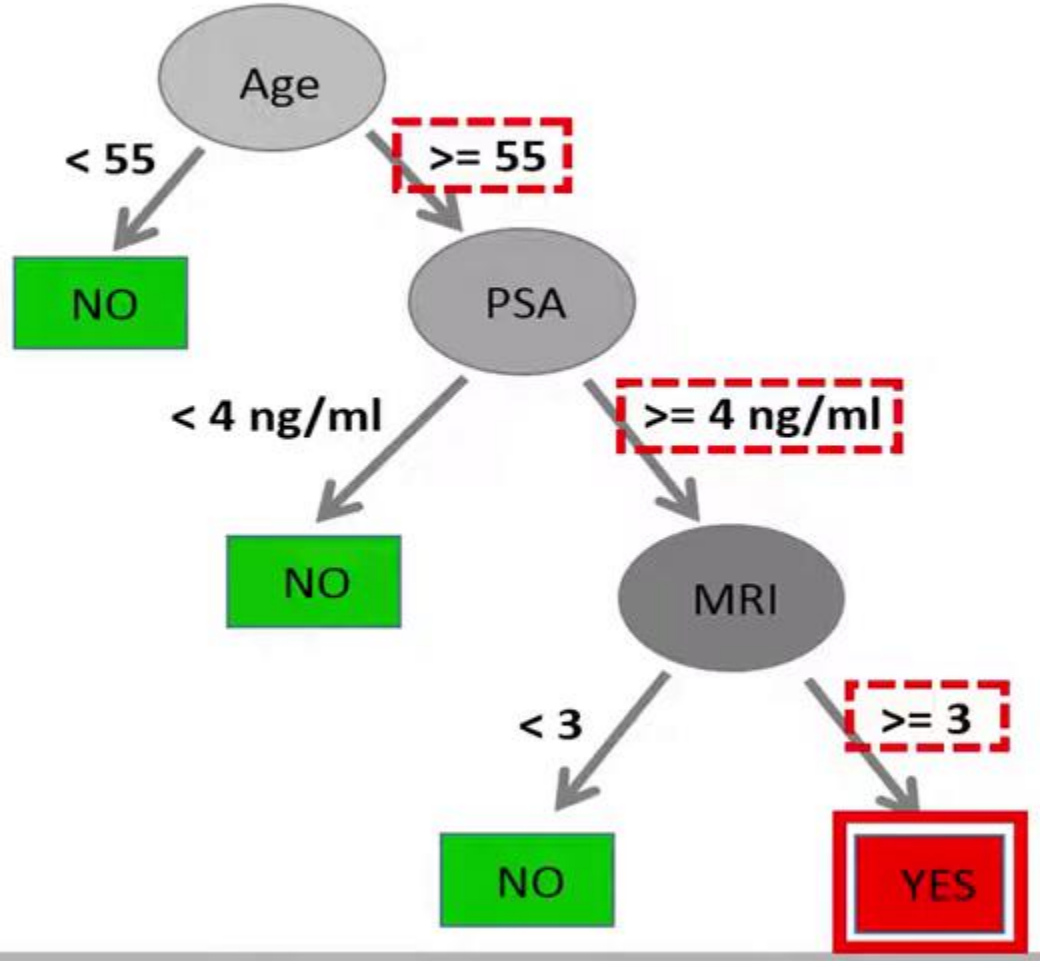


Cancer
Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

and an MRI score that is equal or greater than 3,

Decision tree



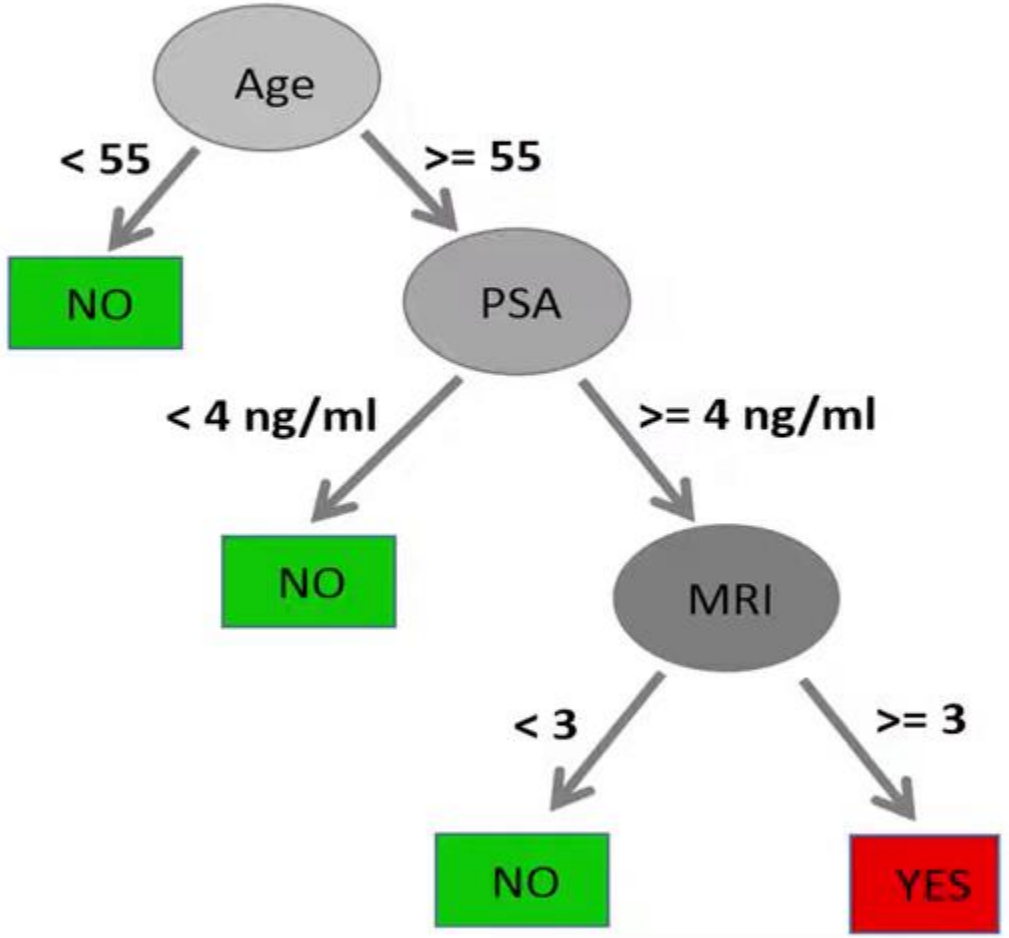
Cancer

Healthy

	Person	Age	PSA	MRI
1	1	85	5.1	3
2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

the person is predicted to have prostate cancer. To confirm this prediction, one usually takes a biopsy of the prostate to identify tumor cells.

Decision tree



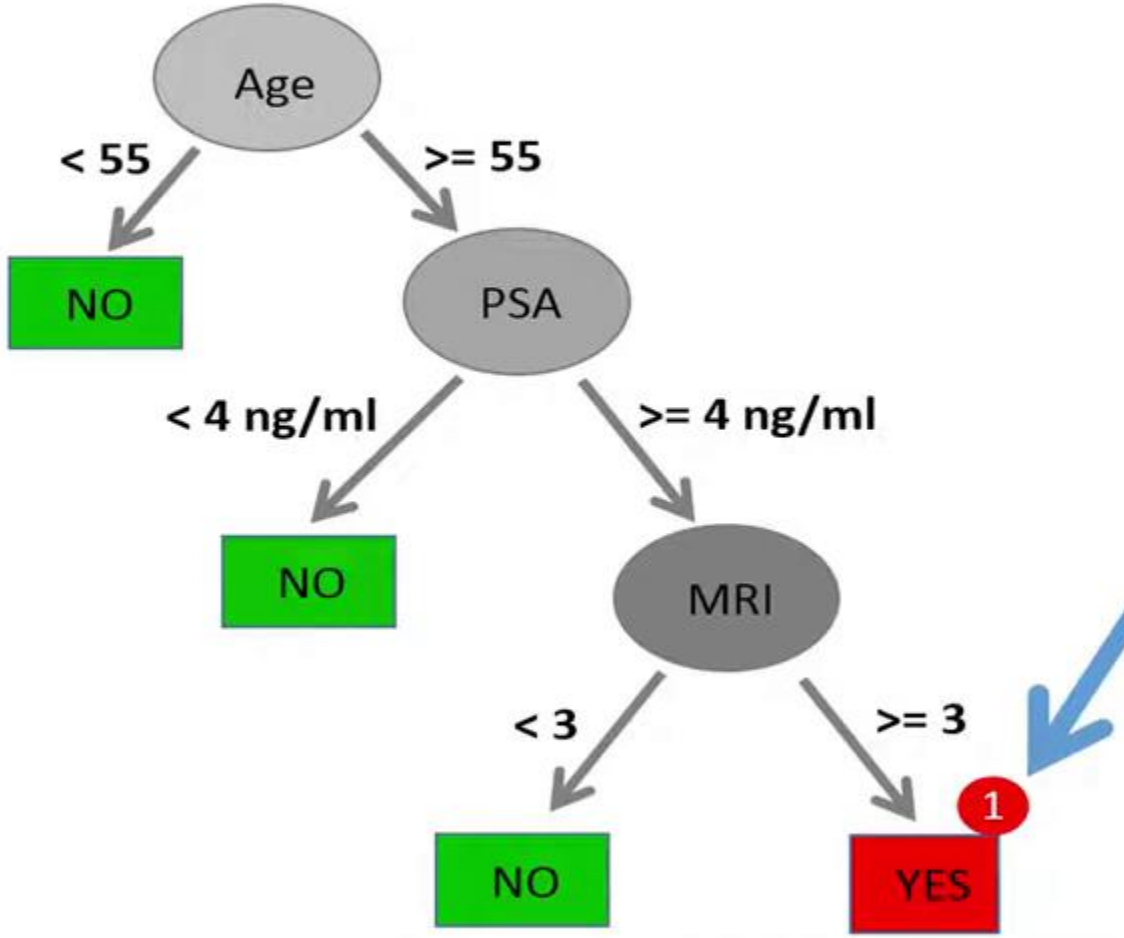
Cancer

Healthy

	Person	Age	PSA	MRI
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2	2	54	4.3	2
3	3	50	1.2	1
4	4	65	4.9	3
5	5	67	5.6	4
6	6	56	6.7	5
7	7	63	6.5	2
8	8	70	3.0	3
9	9	50	4.9	3

This tree was based on the following data.

Decision tree



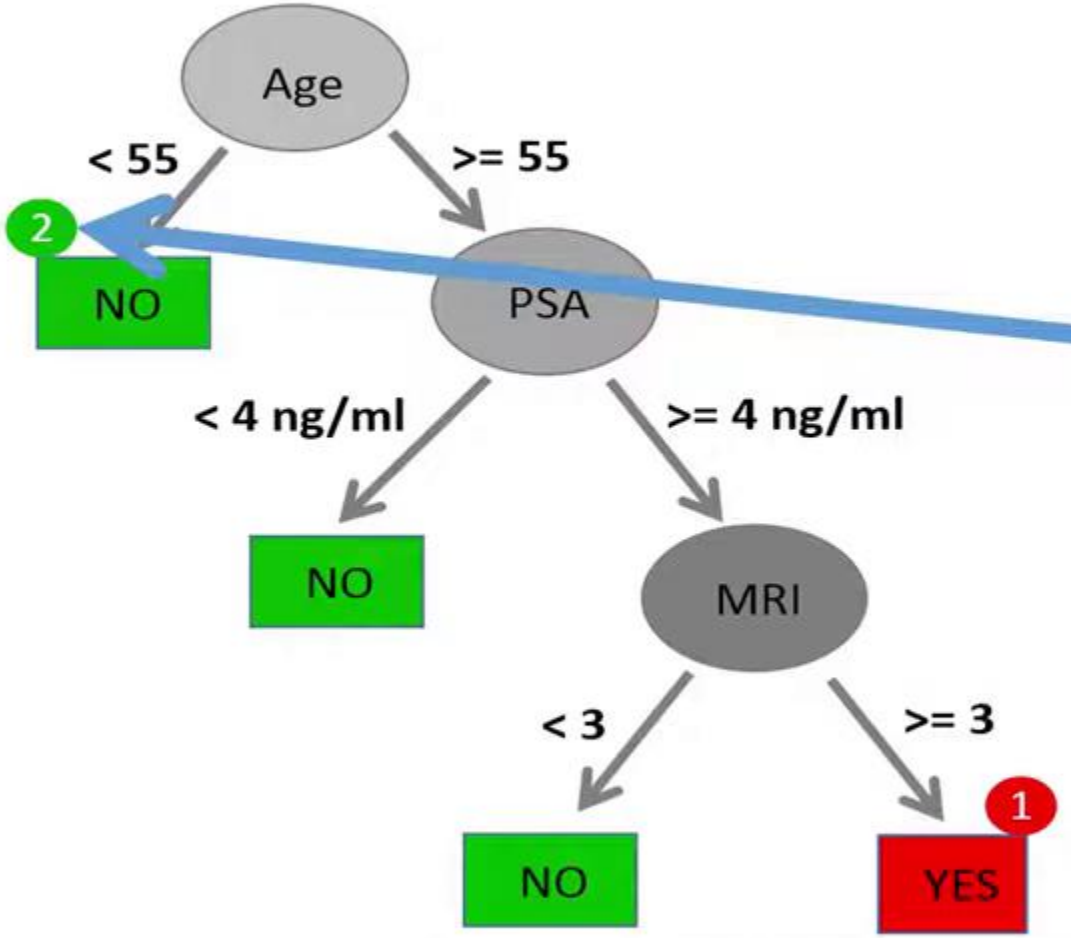
Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

The following person is correctly predicted to have prostate cancer,

Decision tree



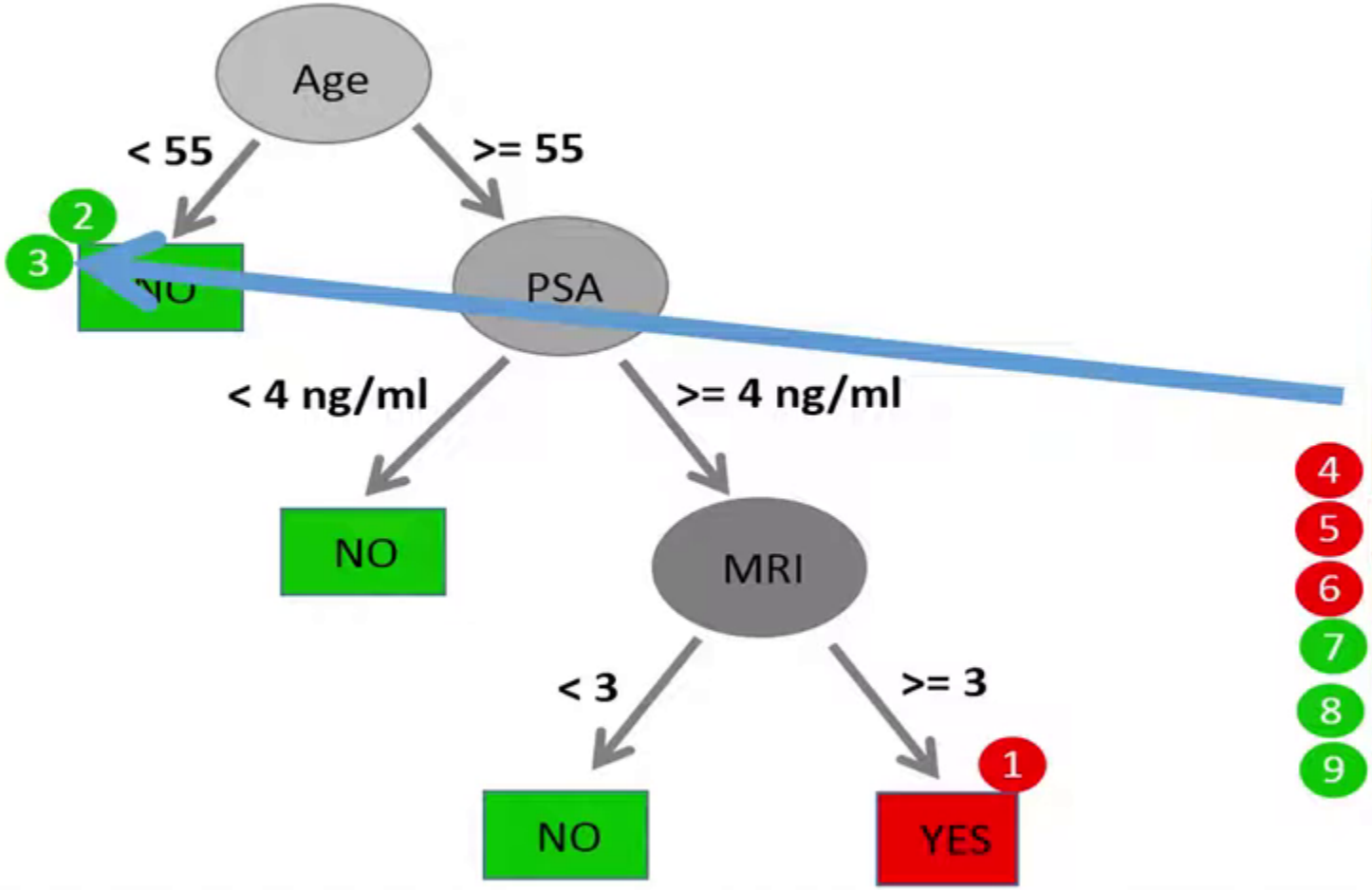
Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

The second person, which we know is healthy, is correctly predicted to be healthy,

Decision tree



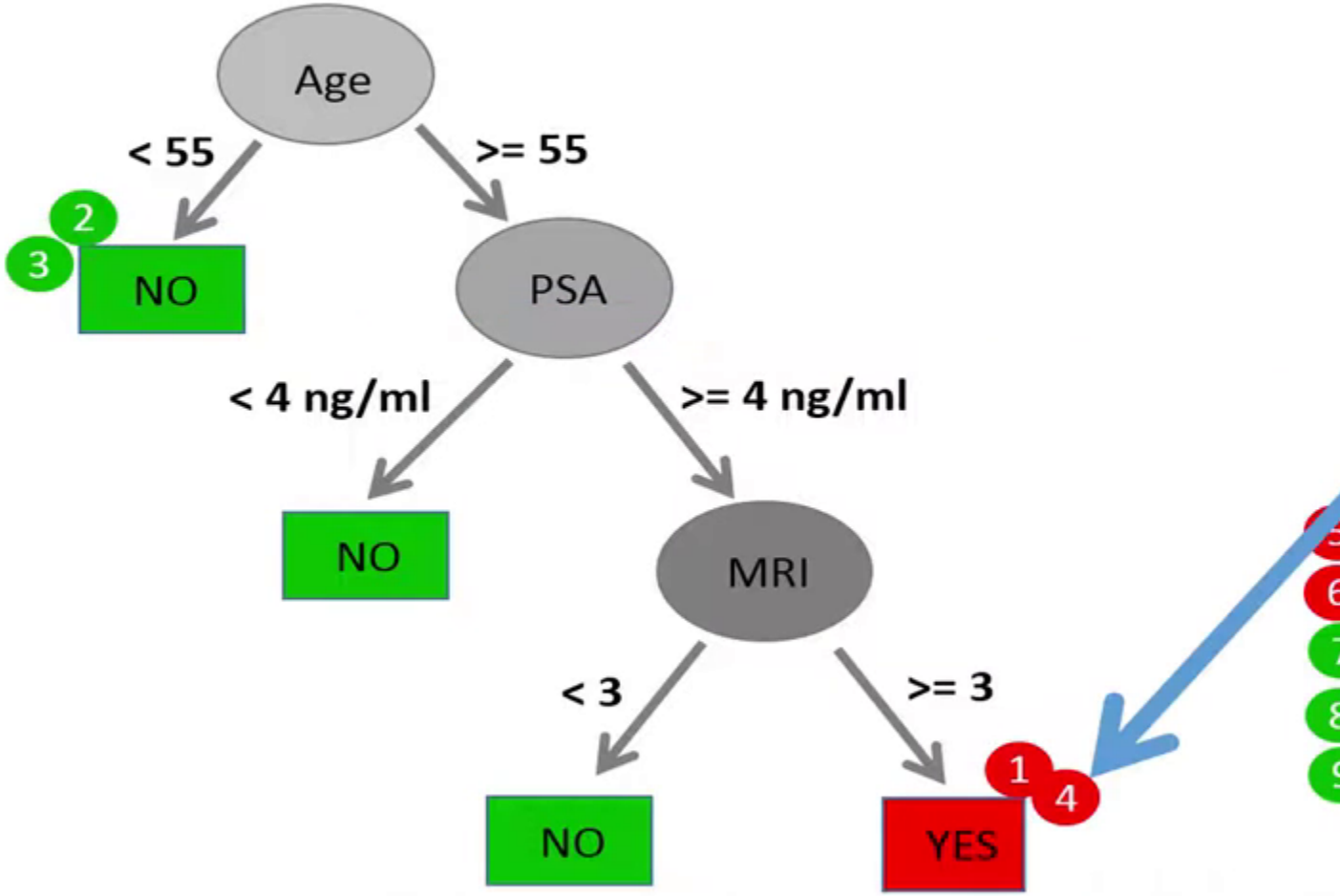
Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

which is also true for person number three.

Decision tree



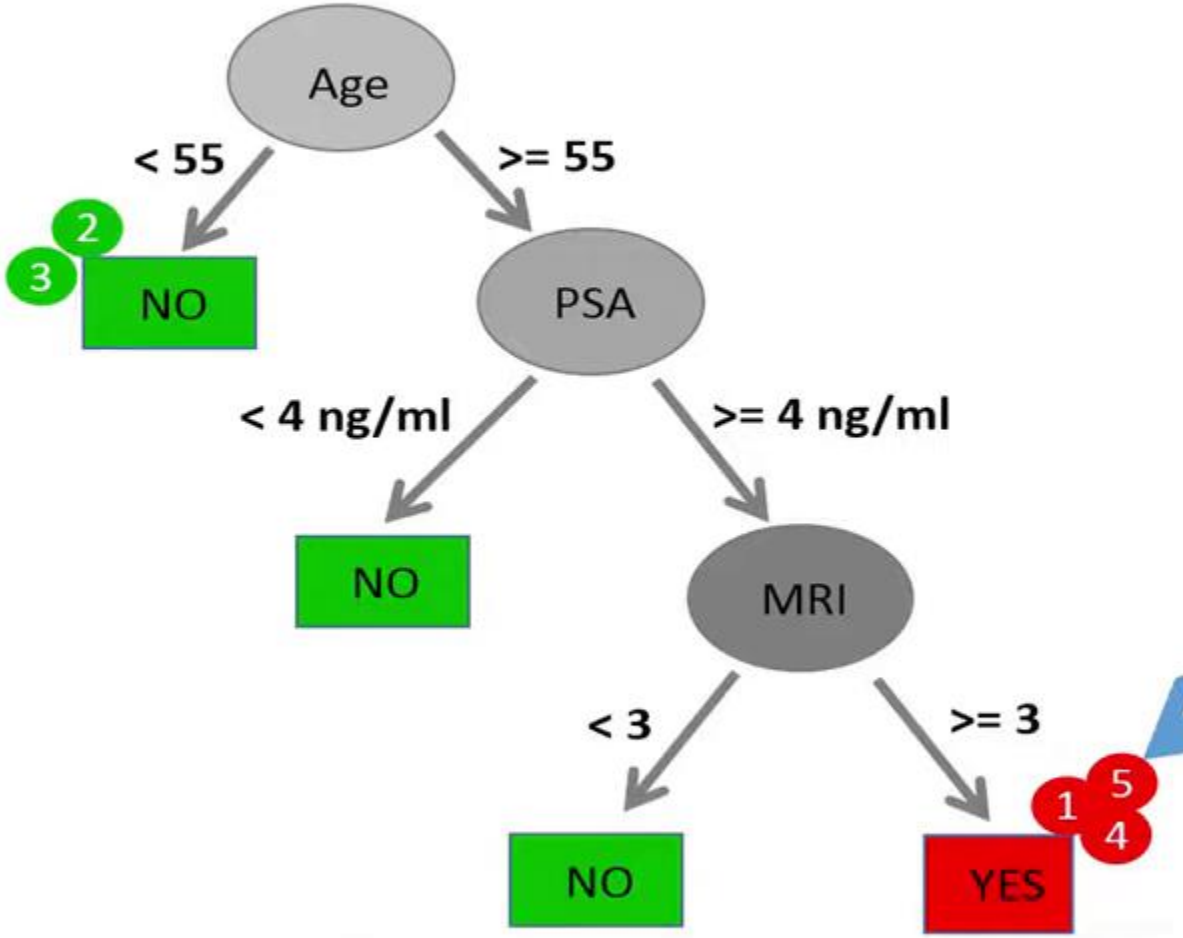
Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

Person number four is correctly predicted to have prostate cancer, and so forth.

Decision tree

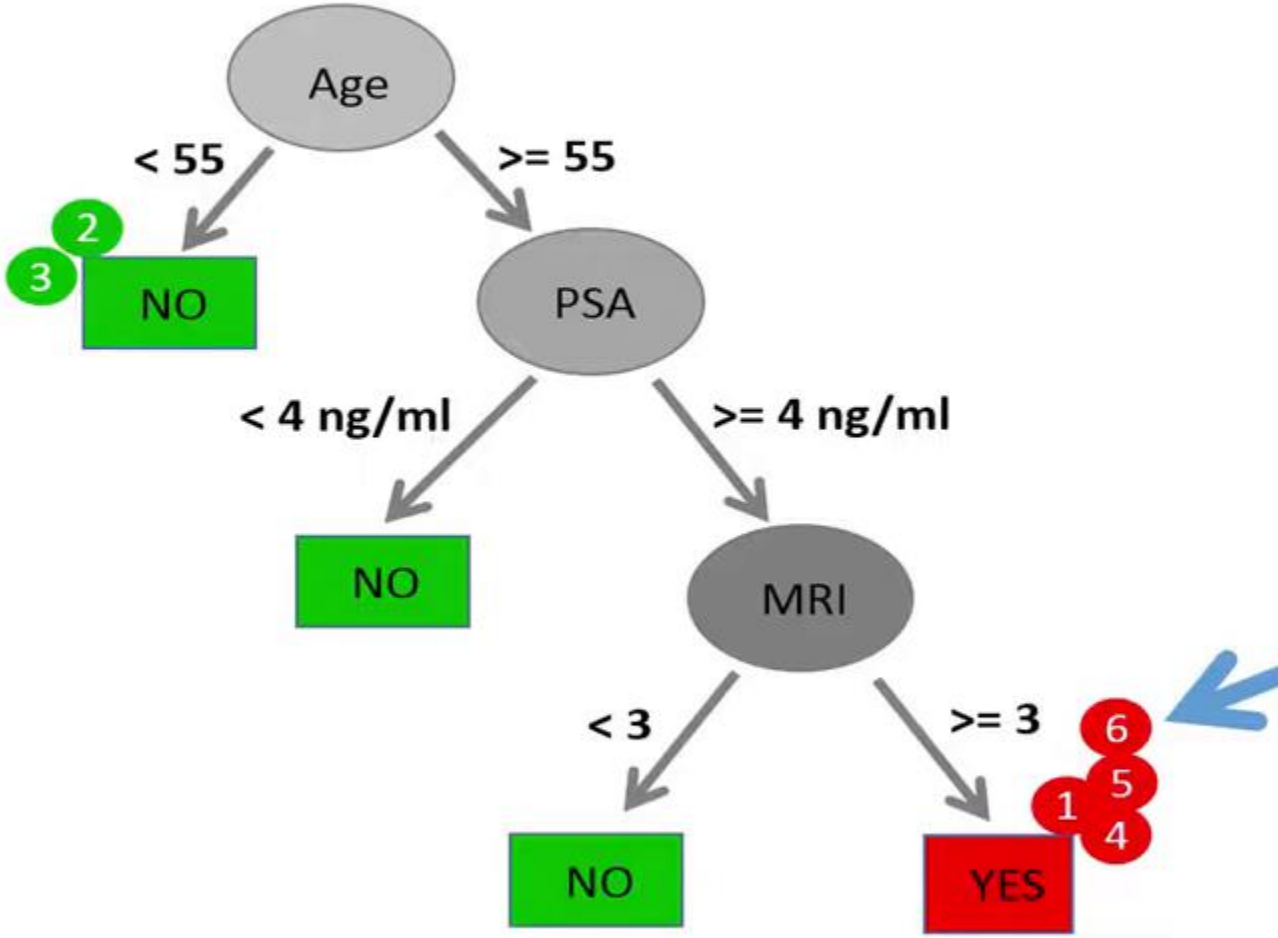


Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

Decision tree

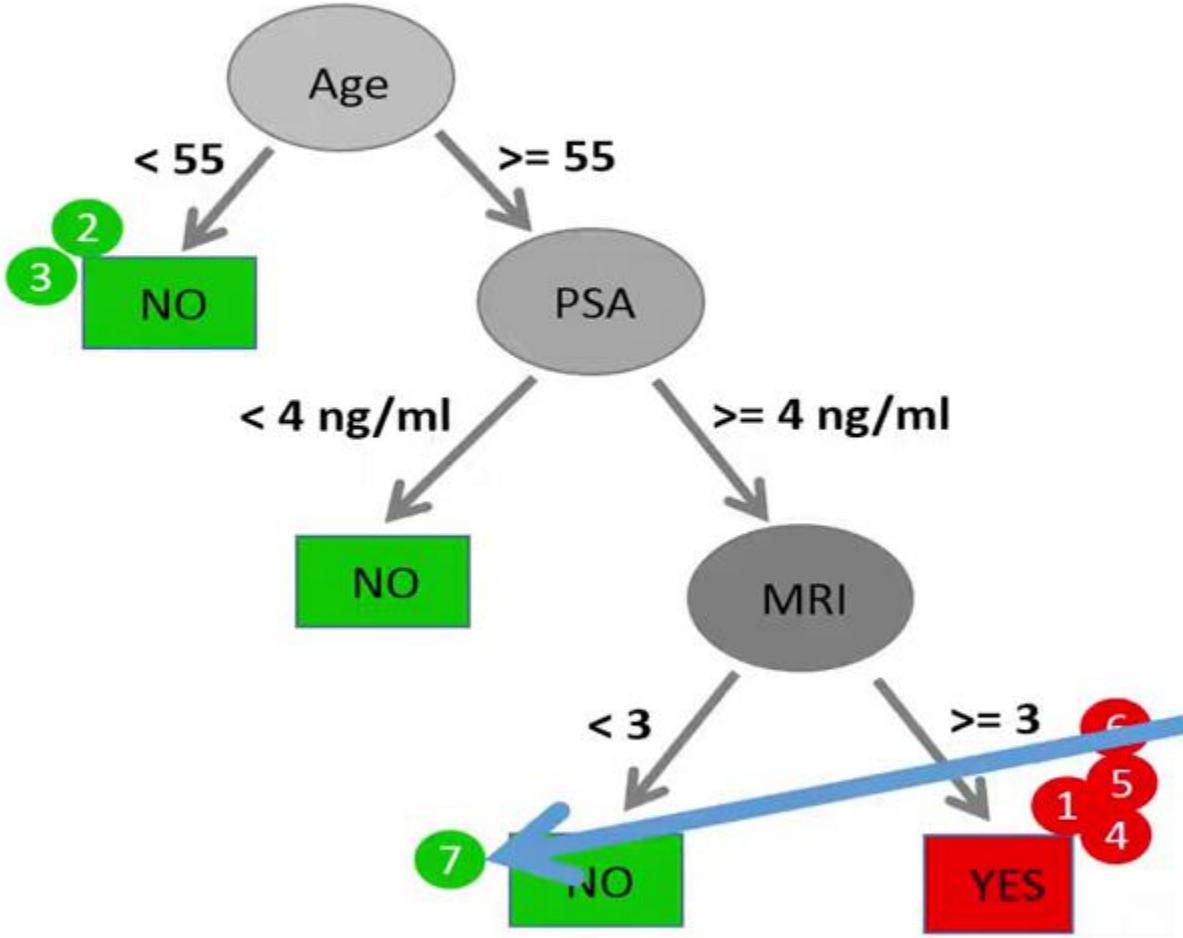


Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

Decision tree

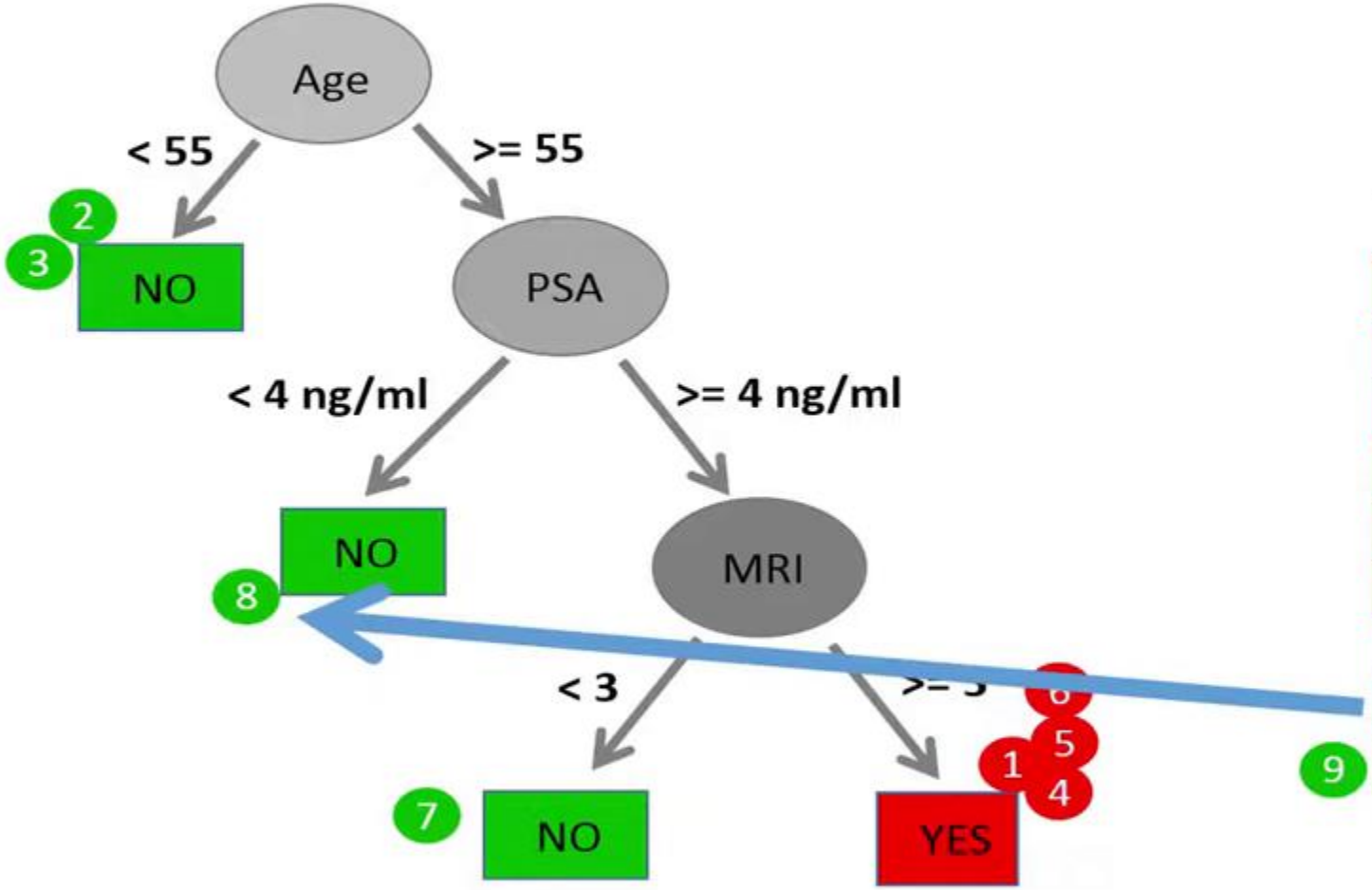


Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

Decision tree

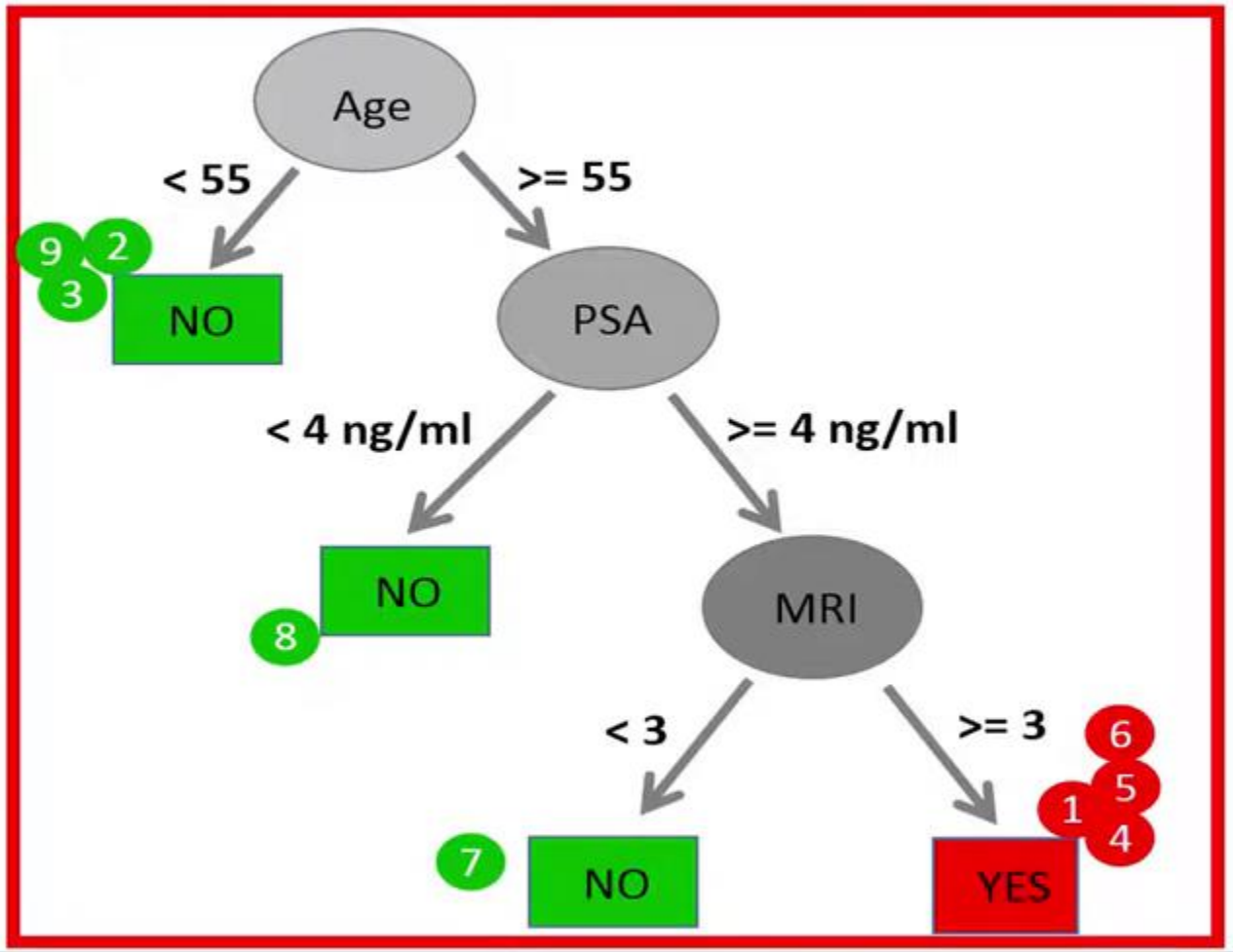


Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

Decision tree



Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

We see that the decision tree makes a perfect prediction because all healthy individuals are predicted to be healthy and all patients with prostate cancer are correctly predicted to have prostate cancer.

Random forest



Random forest

To overcome the problem by overfitting and large variance when using decision trees, the method of random forest was developed.

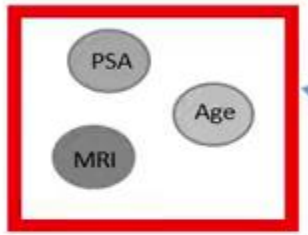
Random forest



Random forest

A random forest is simply a forest of many decision trees, where we base the prediction on many trees, which then reduces the variation and removes the problem of overfitting.

Random forest





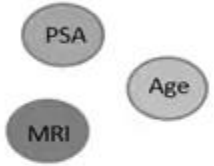
Cancer

Healthy

Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

To explain how a random forest works, let's begin by putting our three variables up here.

Random forest



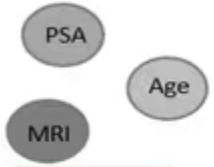




Person	Age	PSA	MRI
1	[illegible]	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

And the nine individuals up here.

Random forest



M = 3



Cancer

Healthy

We let *M* define the number of variables we have, which is here equal to three,

Random forest



M = 3



n = 9

Cancer

Healthy

whereas *n* defines the number of individuals we have, which is equal to nine.

1. Take a bootstrap



M = 3

Bootstrap



n = 9

Cancer

Healthy

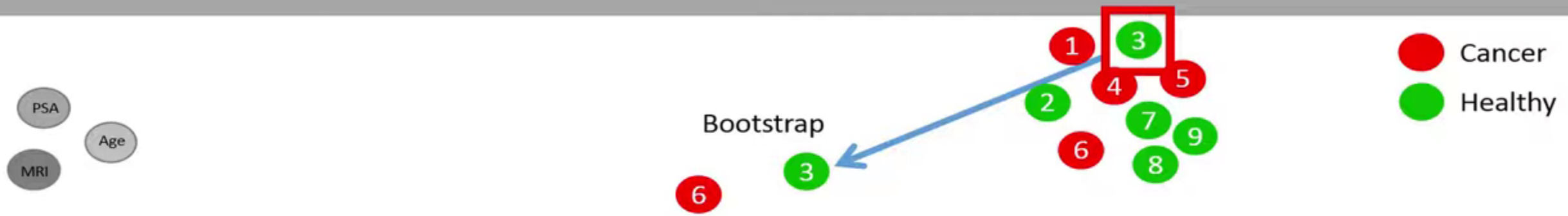
The random forest method begins by taking a bootstrap sample of the data with a sample size of *n*. This means that we should select *n* cases at random, with replacement.

1. Take a bootstrap



For example, just by chance, we happened to select person number six,

1. Take a bootstrap



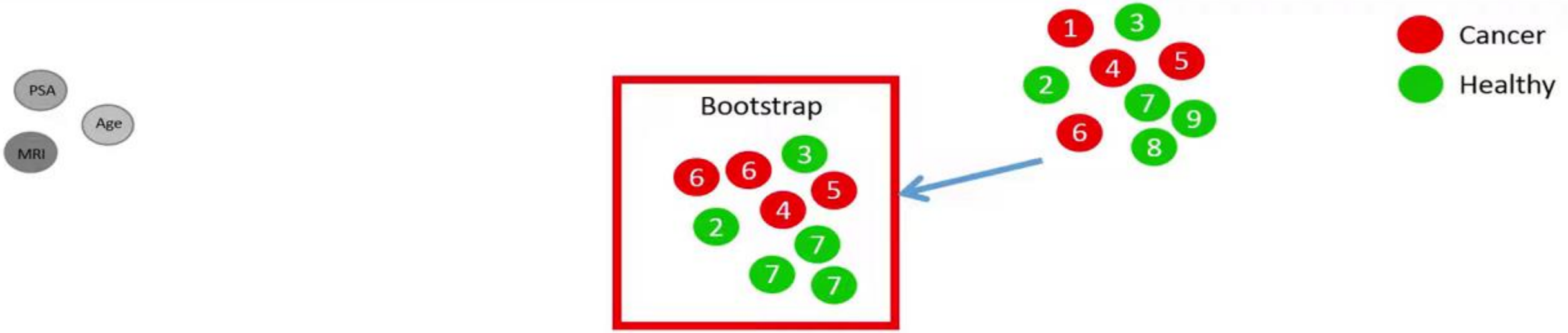
and person number three.

1. Take a bootstrap



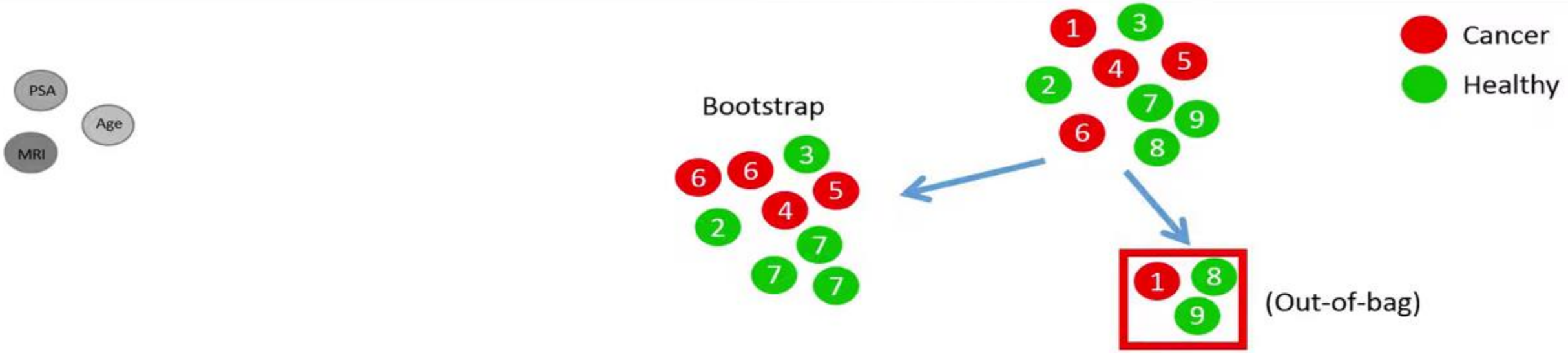
Since we sample with replacement, we might happen to select the same individual again. We here happened to select person number six a second time.

1. Take a bootstrap



We continue like this until we have collected a bootstrap with nine cases, which can be seen as the training data set.

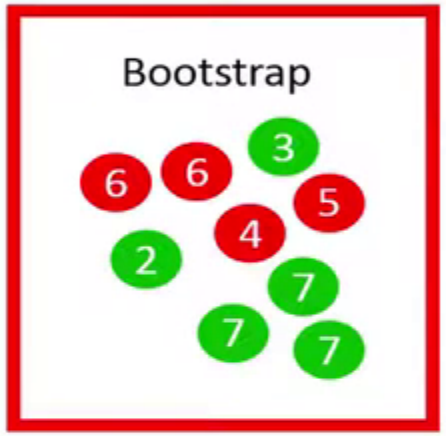
1. Take a bootstrap



The number of out-of-bag cases can vary due to chance but usually represents about one-third of all the cases.

2. Build a tree





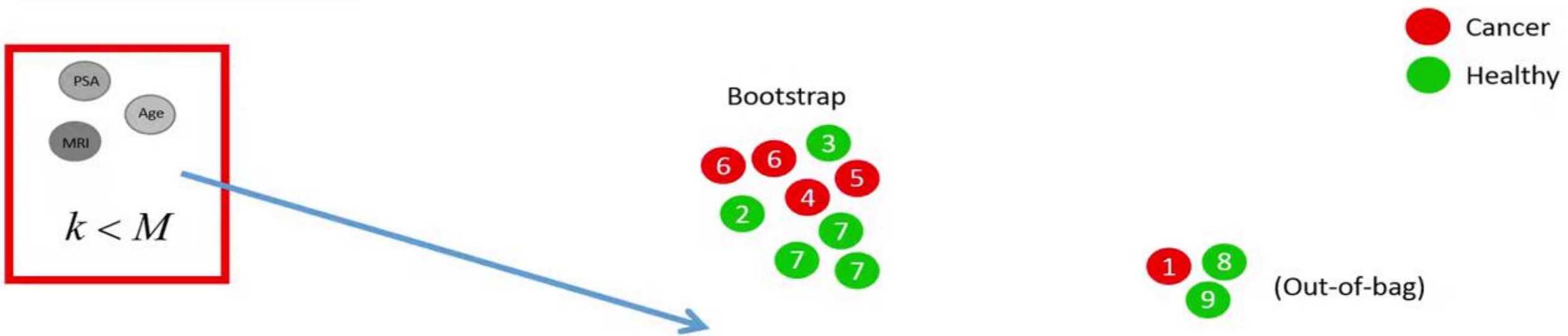




(Out-of-bag)

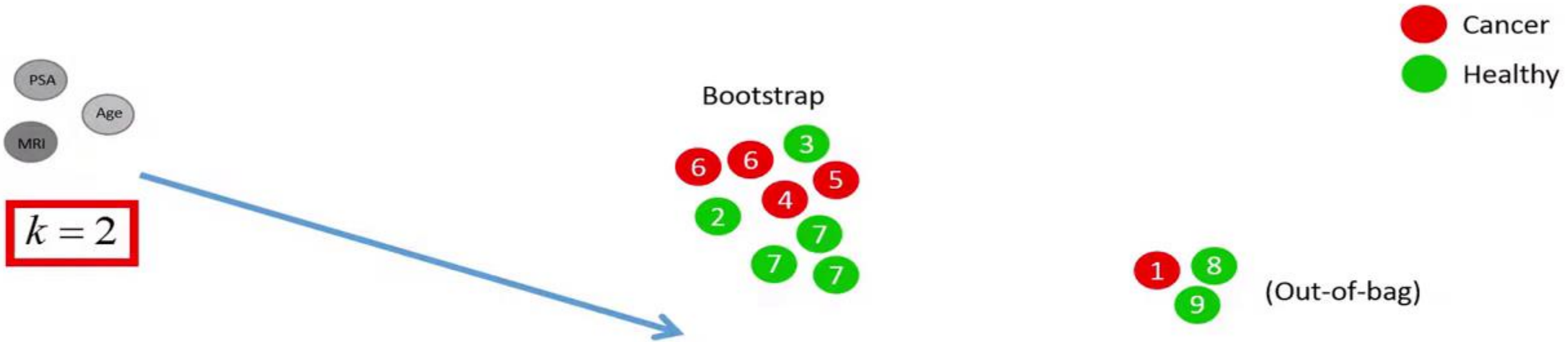
In step two, we build a decision tree based on the bootstrap sample.

2. Build a tree



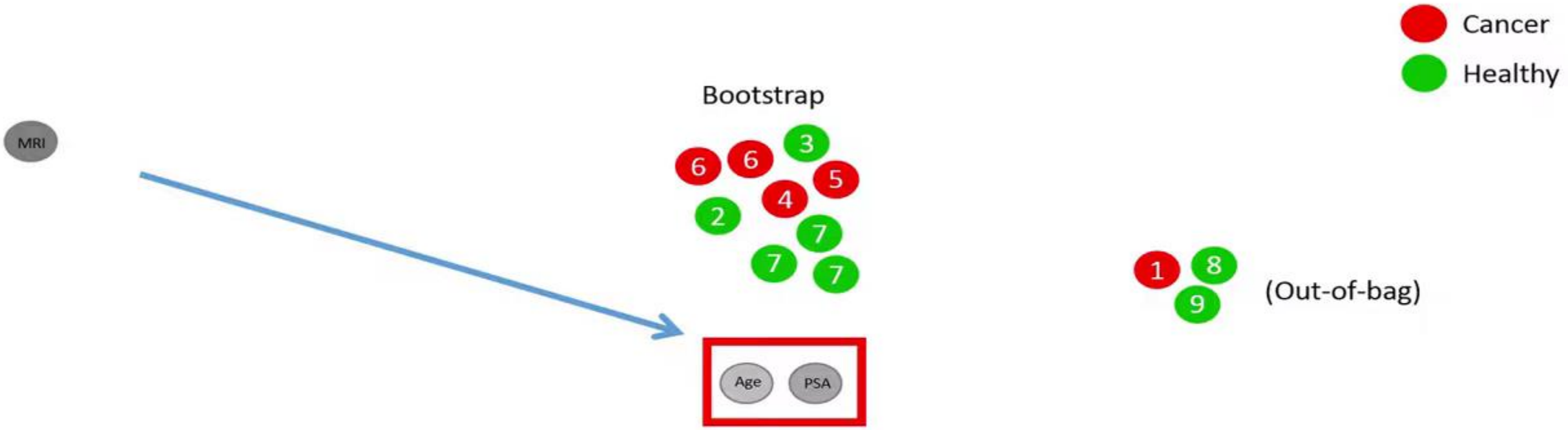
However, at each node, we only randomly select *k* variables (or features) out of the total number of variables to generate a variety of trees.

2. Build a tree



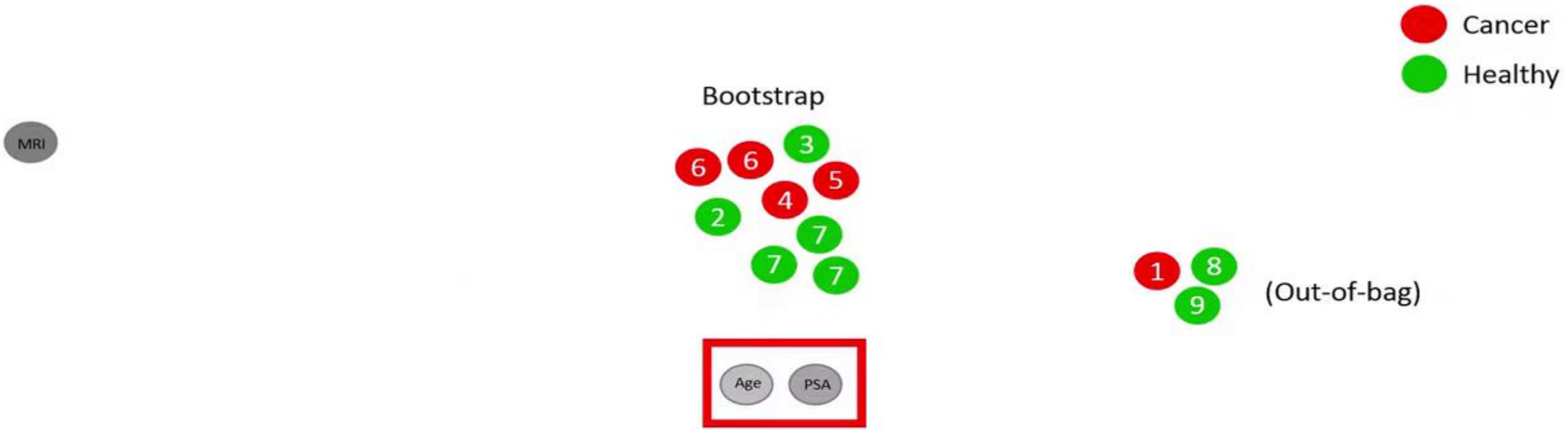
Usually, a recommended value of *k* should be equal to the square root of the number of variables. In this example, we therefore set *k* equal to two.

2. Build a tree



This means that we should select two out of the three variables. For example, let's say that the two variables age and PSA were selected.

2. Build a tree

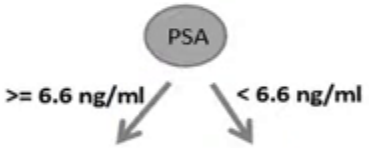


The variable that results in the lowest Gini index, , is then selected as the root node.

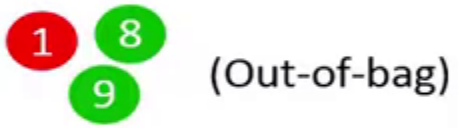
2. Build a tree





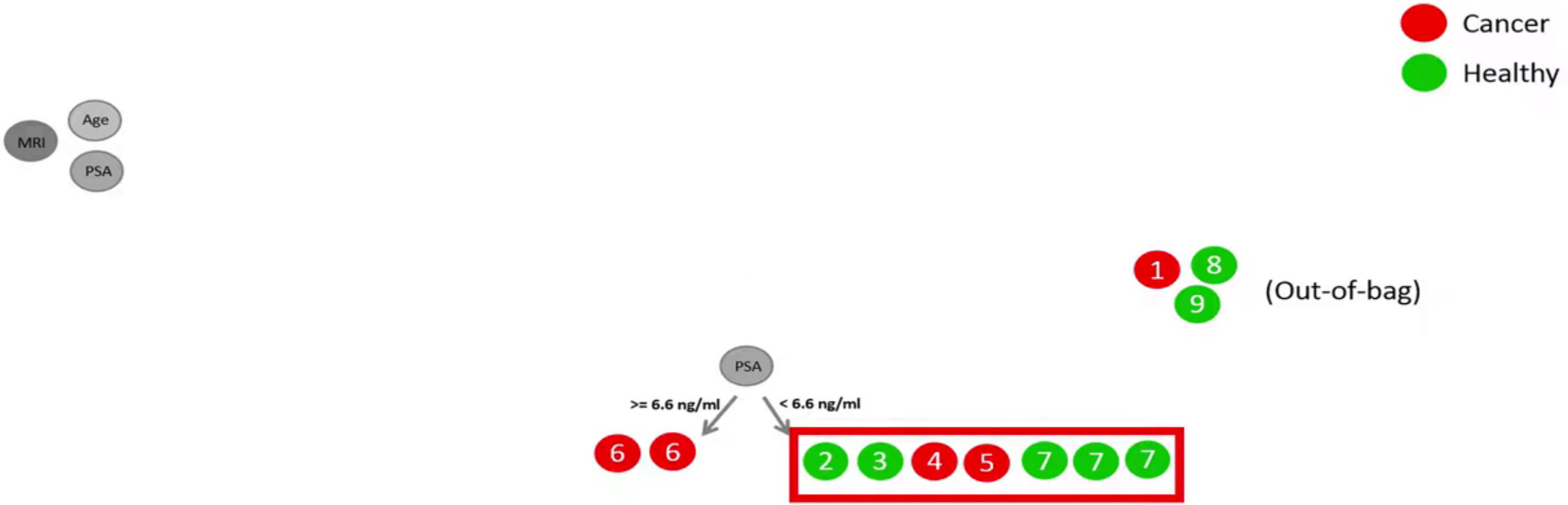






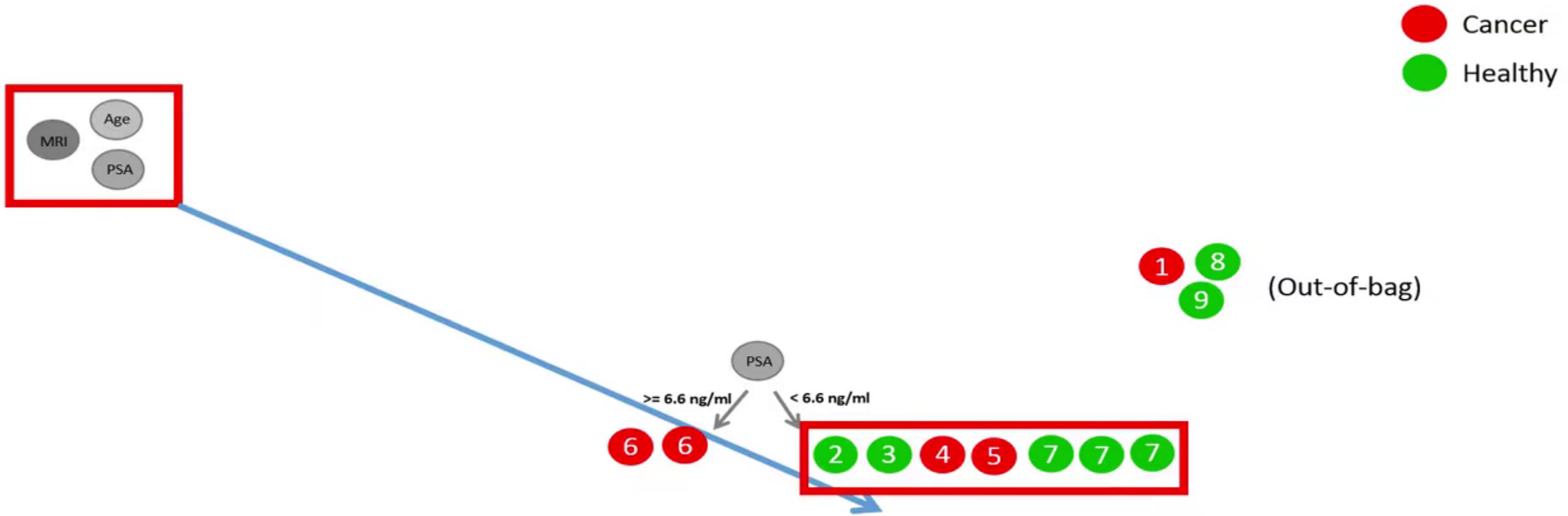
Suppose that the variable PSA resulted in the lowest Gini index based on the bootstrap sample.

2. Build a tree



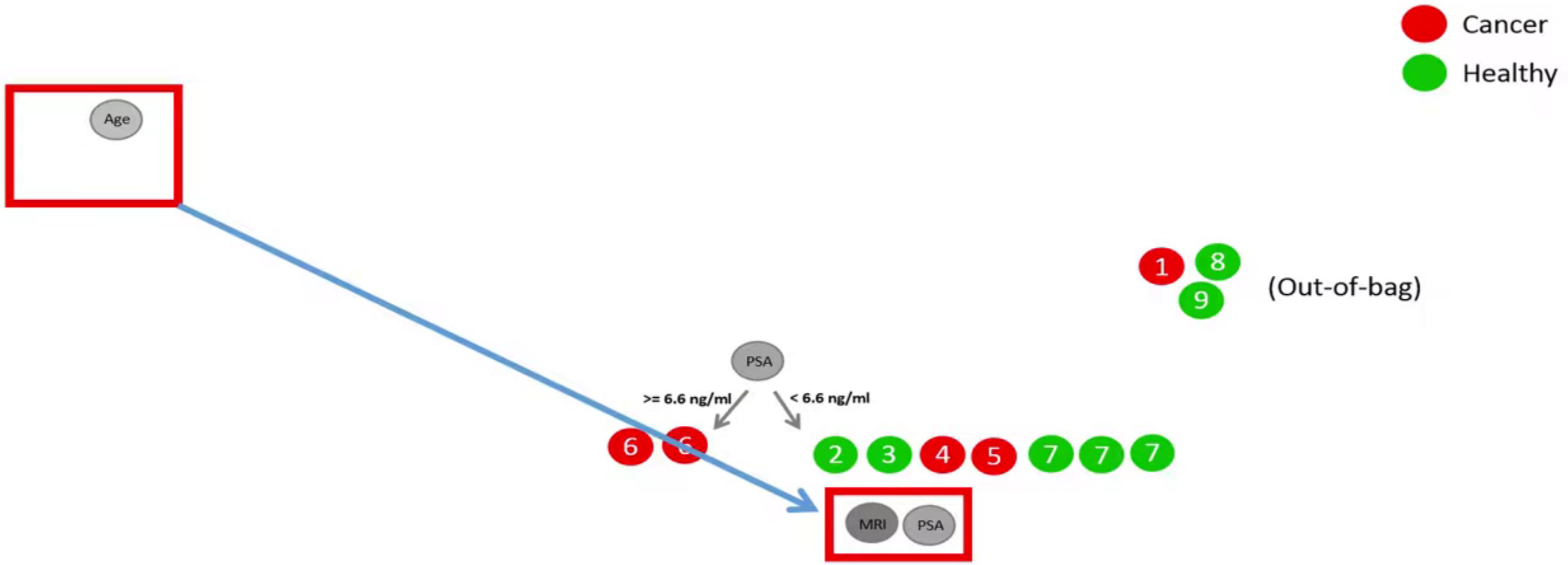
We will now continue to grow the tree until the two groups have been fully separated. We see that we should continue to grow the tree because these data points consist of a mixture of healthy individuals and patients with prostate cancer.

2. Build a tree



We therefore again randomly select two variables.

2. Build a tree

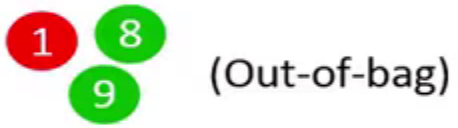


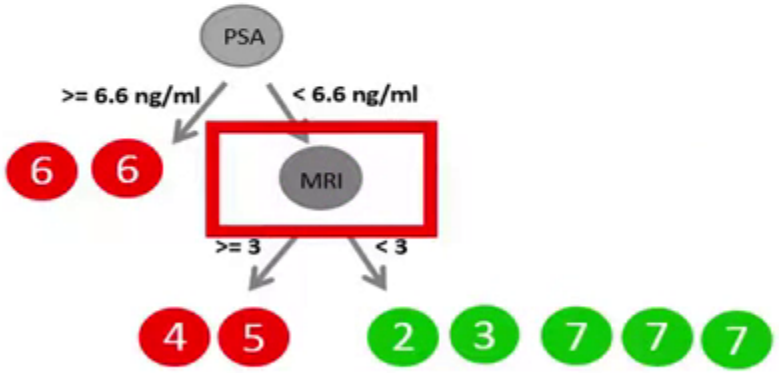
Let's say that we selected PSA and MRI this time. Suppose that MRI results in the lowest Gini index out of these two variables.

2. Build a tree







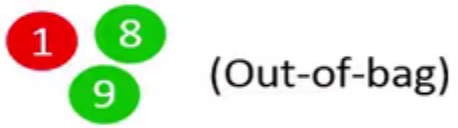


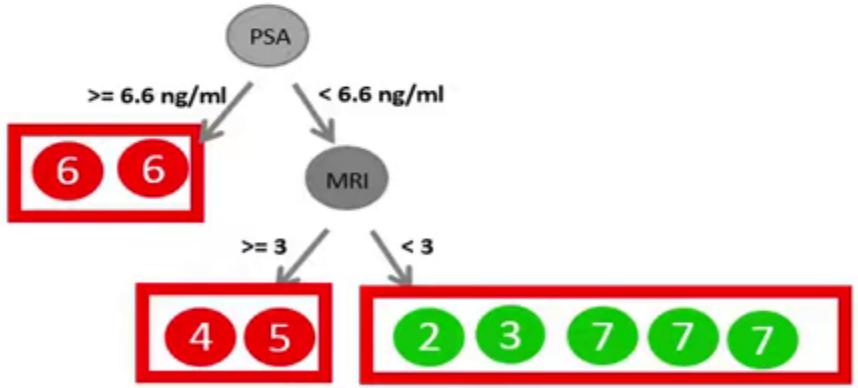
The variable MRI is therefore selected as the second node in the tree.

2. Build a tree









Since the leaf nodes are now completely pure, the tree stops growing.

2. Build a tree

Cancer

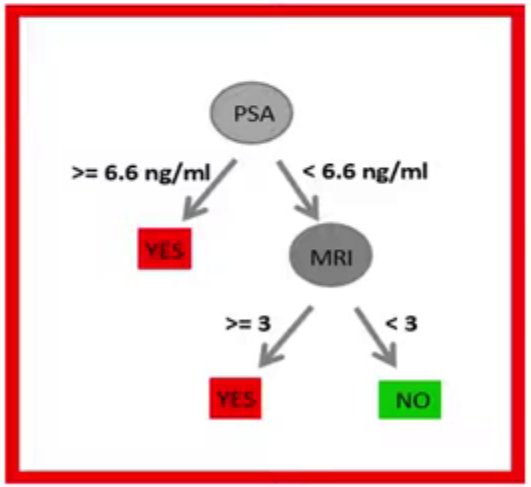
Healthy

Age

MRI

PSA

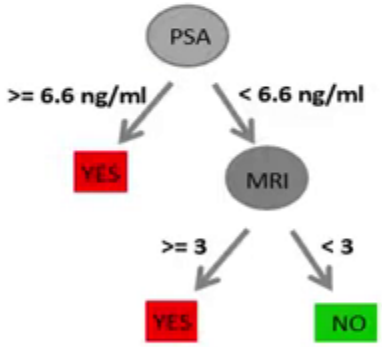
1 8 9 (Out-of-bag)



The final tree looks like this.

Validate tree







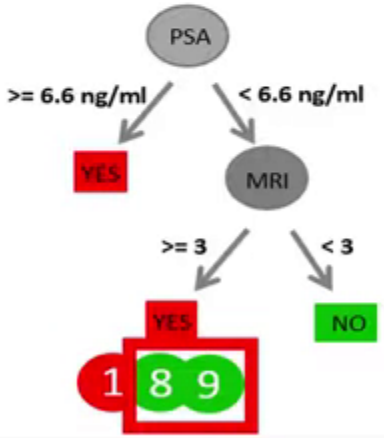
Cancer

Healthy

We can now validate how well this tree predicts on the out-of-bag data.

Validate tree





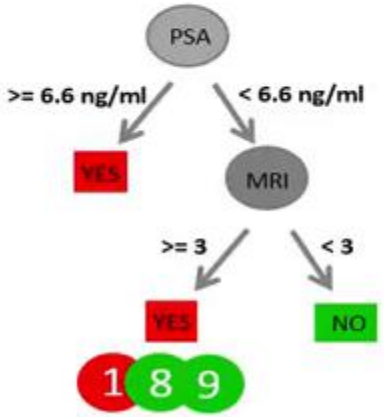
Cancer

Healthy

whereas person number eight and nine were incorrectly predicted by the tree to have prostate cancer.

Validate tree







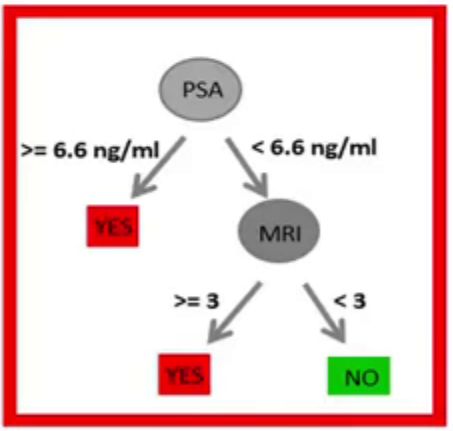
$$\text{OOB error} = 2 / 3 \approx 66\%$$

Since two out of three individuals were incorrectly predicted, the so-called out-of-bag error is therefore equal to about 66% for this tree.

Cancer

Healthy

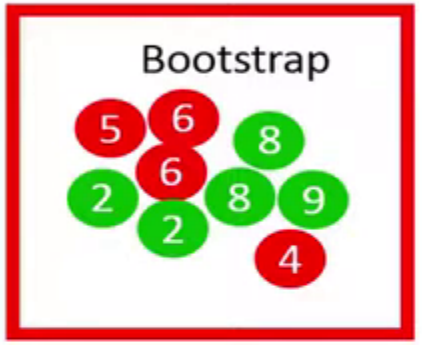


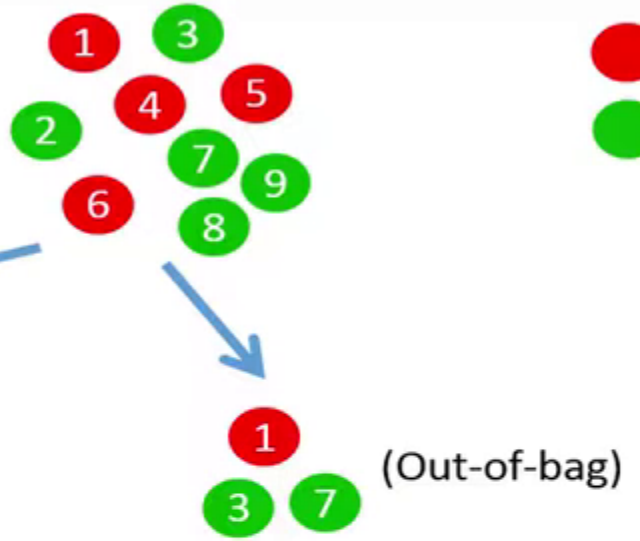


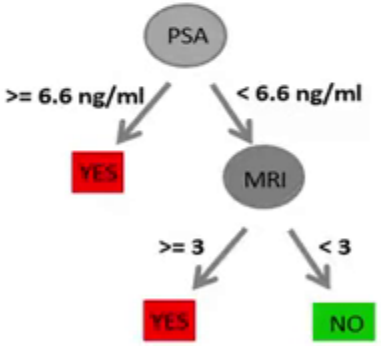
Let’s put aside this tree here.

1. Take a bootstrap









We now repeat the first two steps again, where we take a new bootstrap of the same size as before.

1. Take a bootstrap

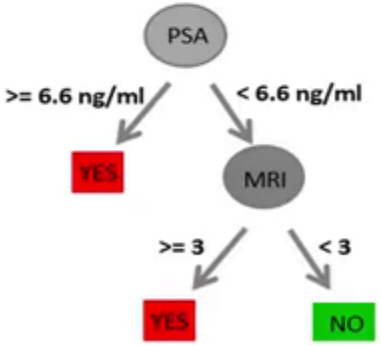






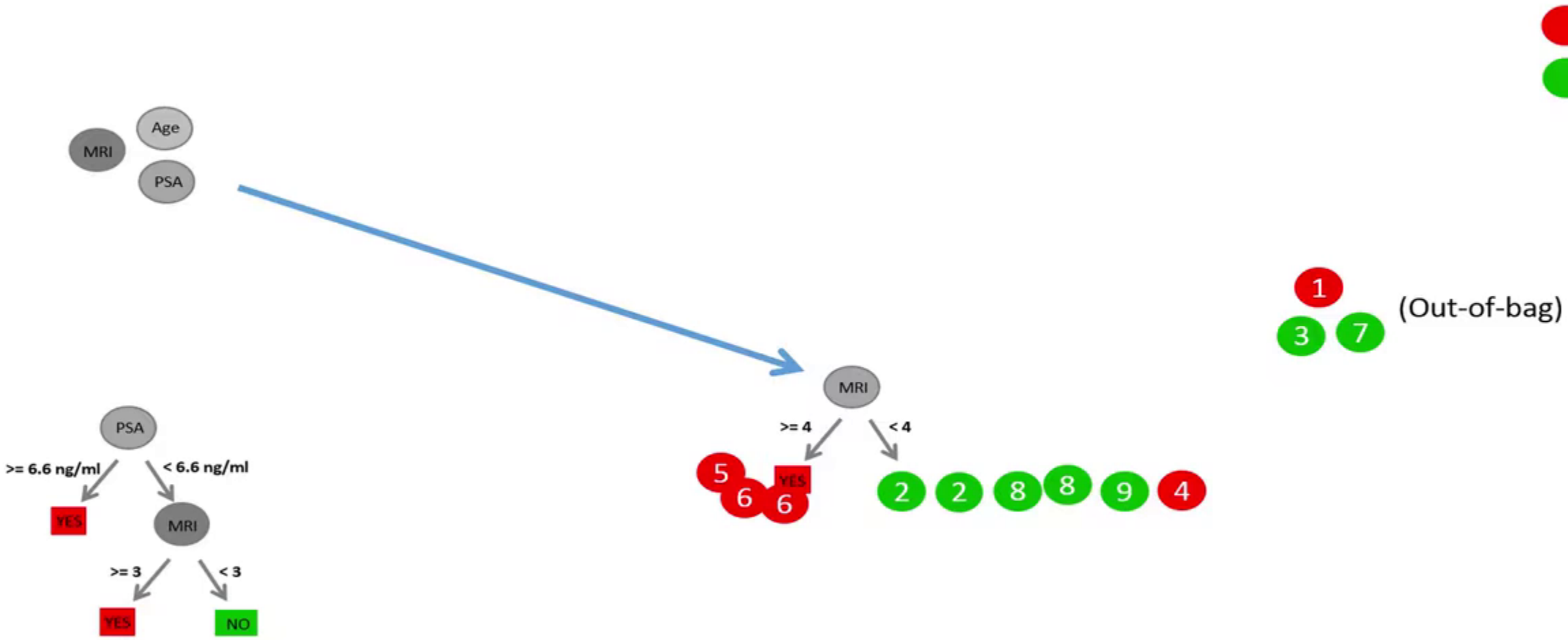






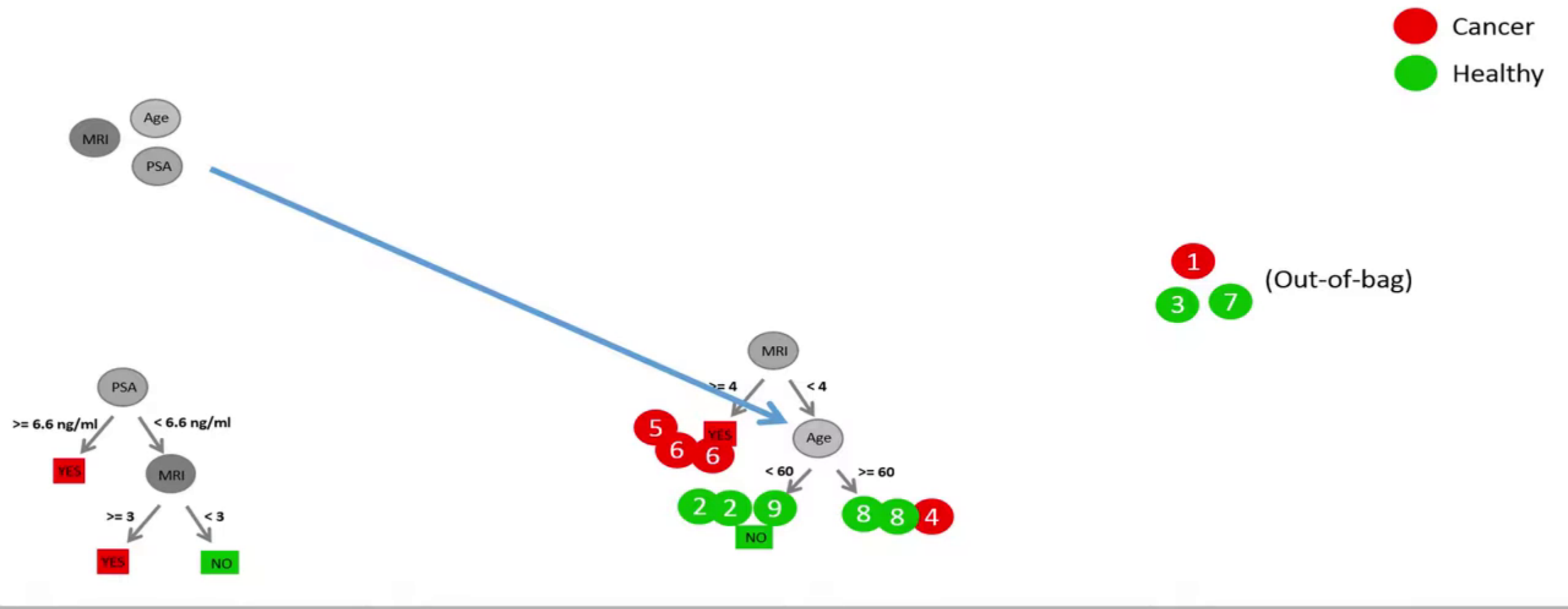
This time, person number one, three and seven were, by chance, not included in the bootstrap sample.

2. Build a tree



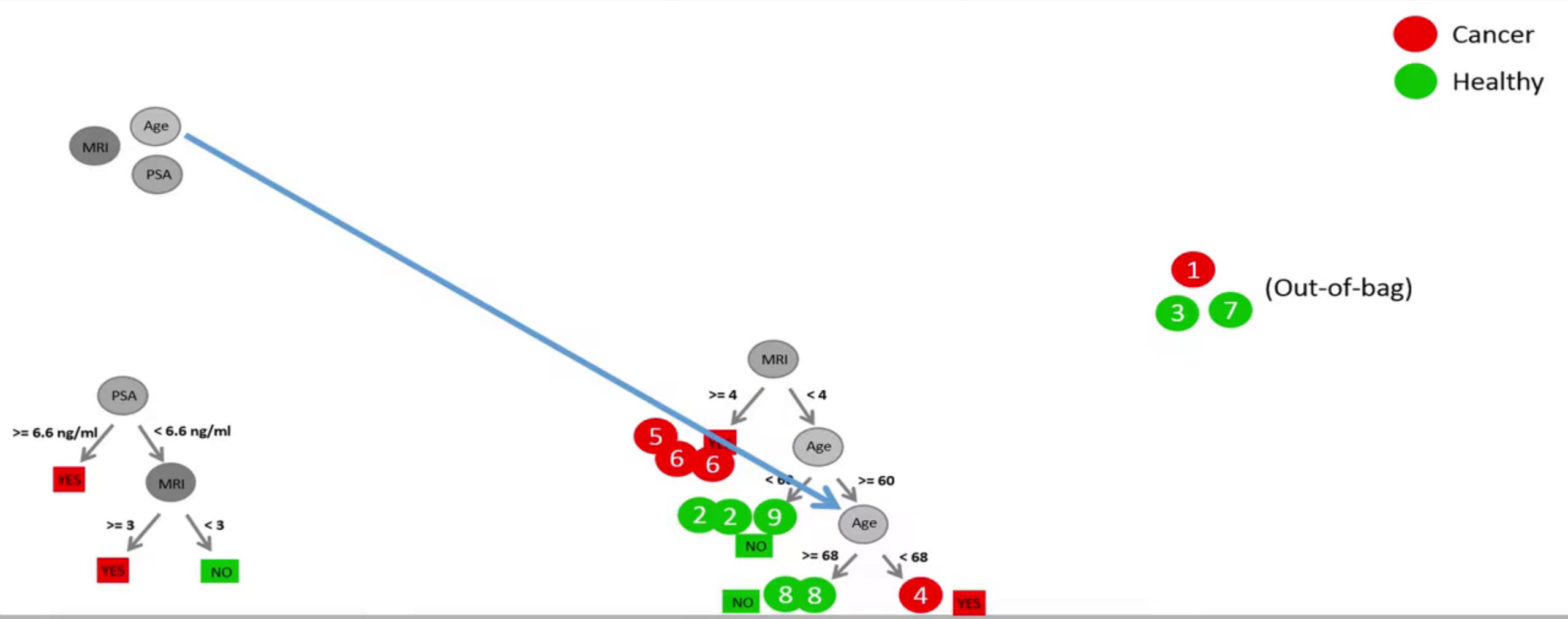
Out of the two randomly selected variables, the MRI variable generated the best split, which explains why it is here used as the root node.

2. Build a tree



Then we randomly select two new variables, where the variable age now happened to result in the best split.

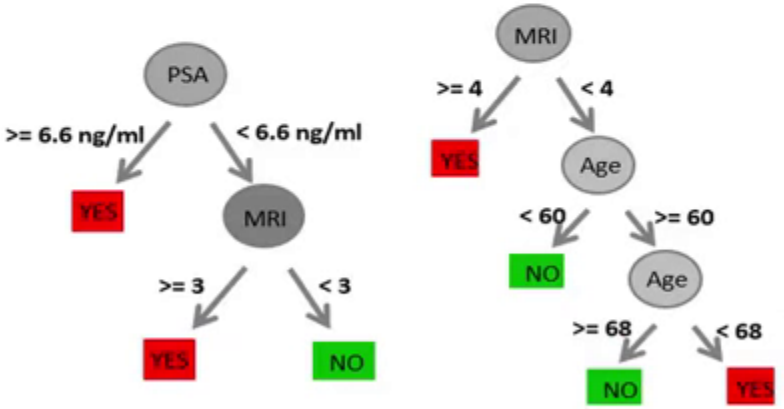
2. Build a tree



We then randomly selected two new variables, where the variable Age again happened to result in the lowest Gini index.

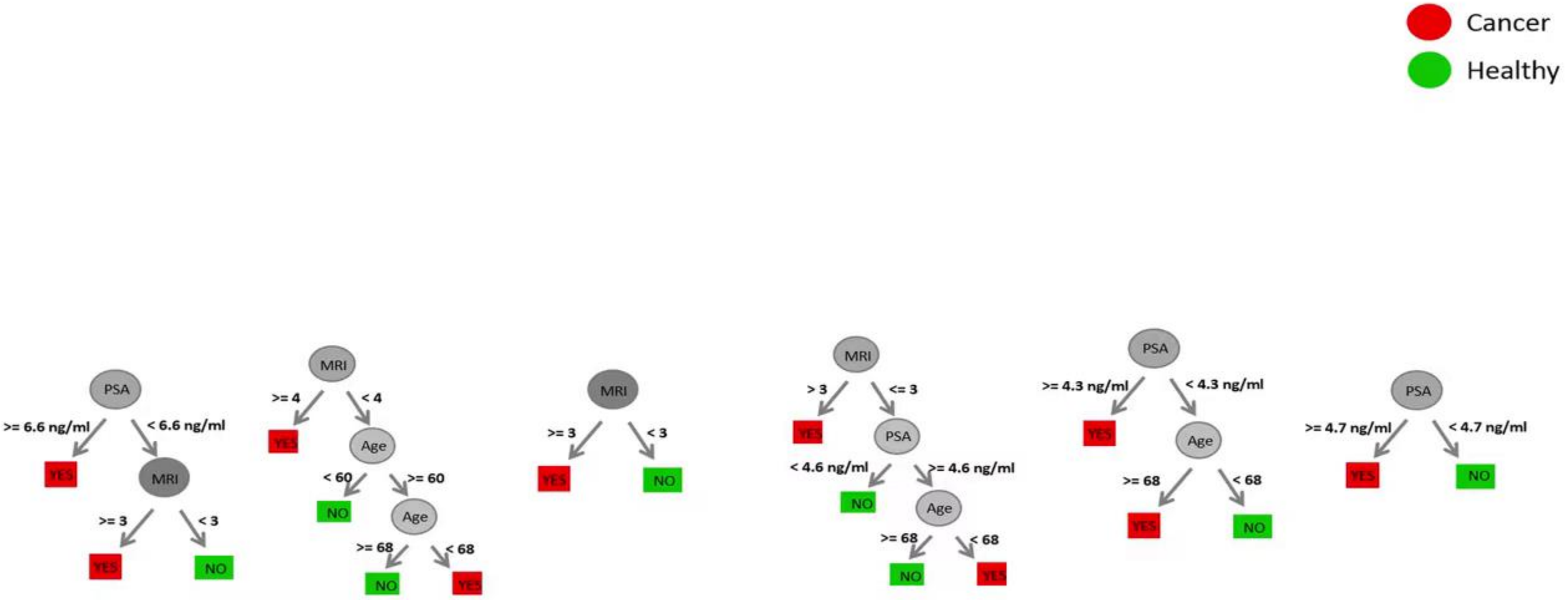
Cancer

Healthy



We now have two trees.

Estimate out-of-bag error



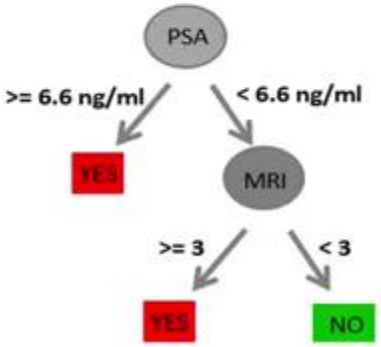
If we continue this process, we will generate many decision trees that form a forest. Let's say that our forest consists of only six trees. Normally, you should have at least a few hundred trees in your forest. The more the better.

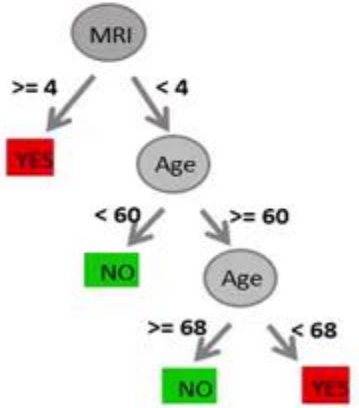
Estimate out-of-bag error

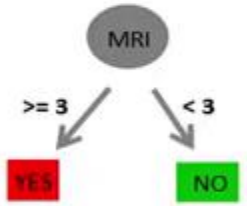
Person	Age	PSA	MRI
1	85	5.1	3
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

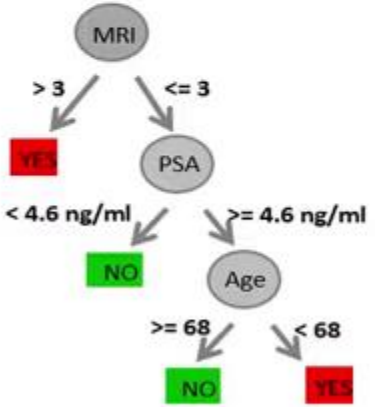
Cancer

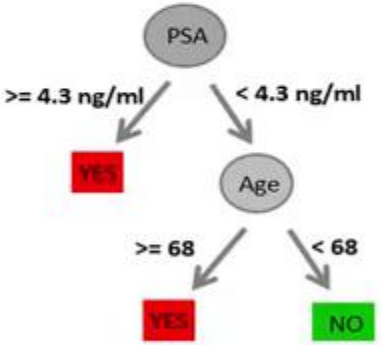
Healthy

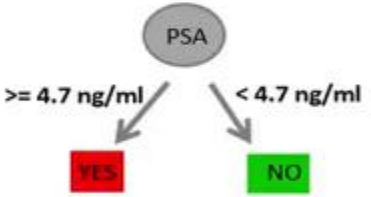












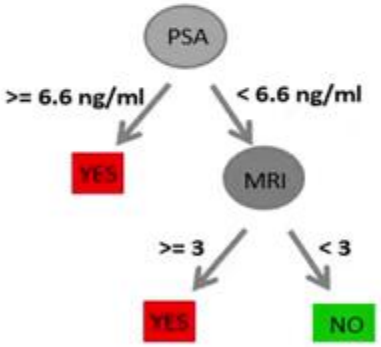
We will now see how this forest predicts each case. The case is predicted based on the majority vote of the trees.

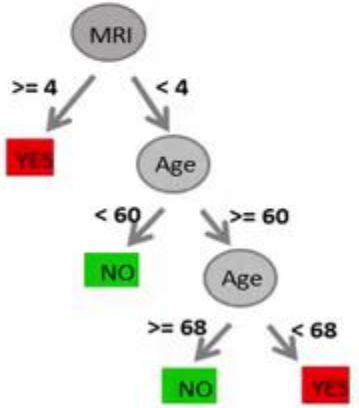
Estimate out-of-bag error

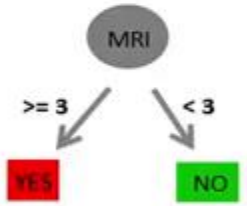
Person	Age	PSA	MRI
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7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

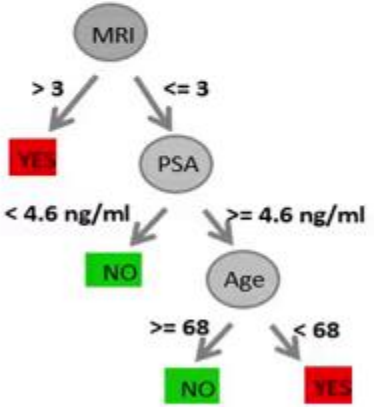
Cancer

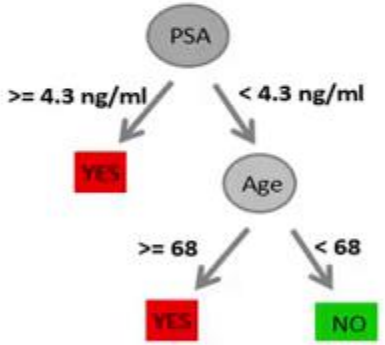
Healthy

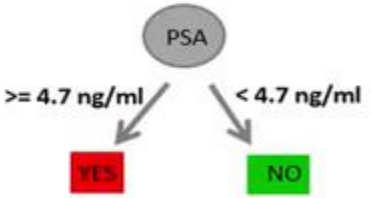












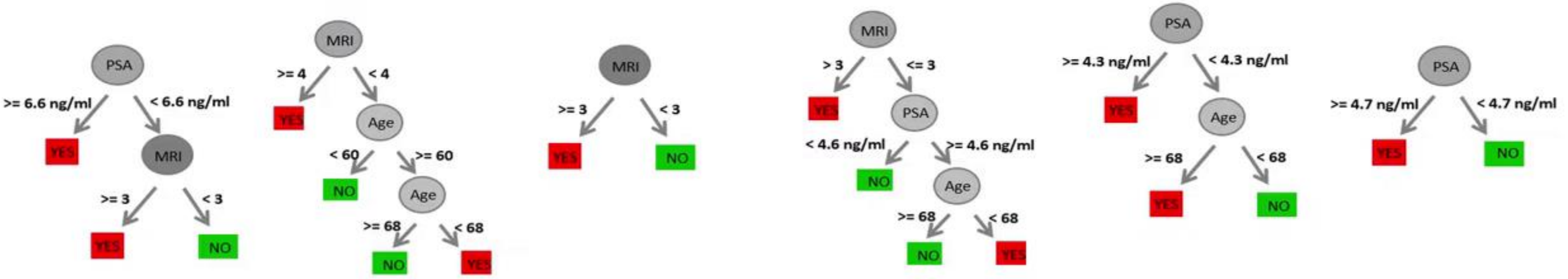
However, only trees that were developed or trained without the influence of the case are allowed to vote. Usually, one third of the trees will therefore vote for each case.

Estimate out-of-bag error

Person	Age	PSA	MRI
1	85	5.1	3
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3	50	1.2	1
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8	70	3.0	3
9	50	4.9	3

Cancer

Healthy



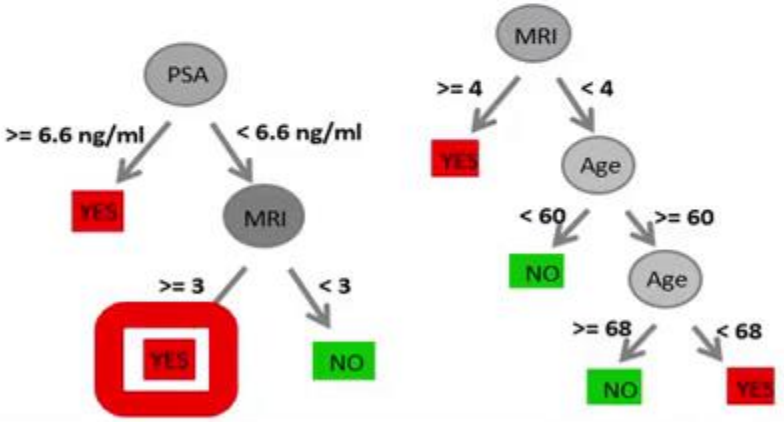
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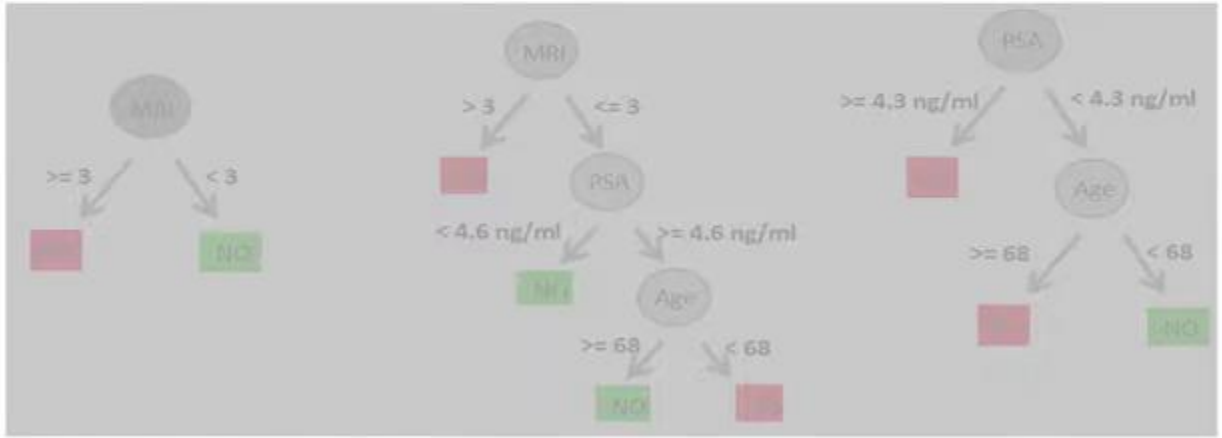
Estimate out-of-bag error

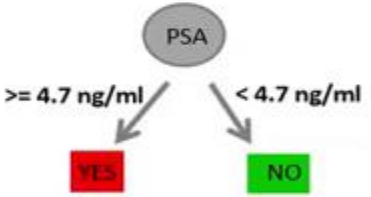
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Cancer

Healthy







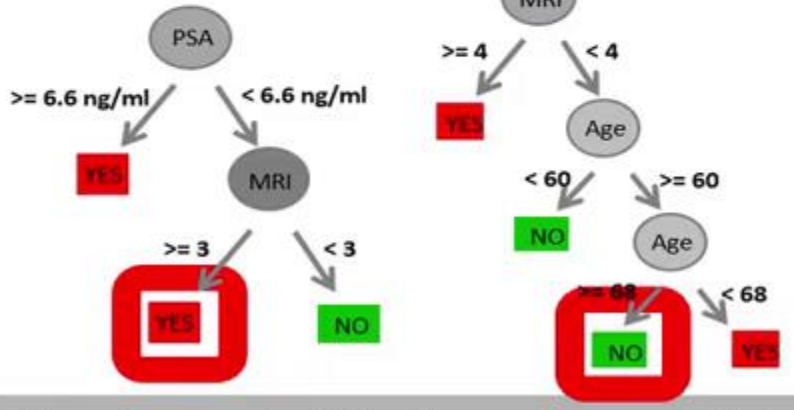
the case is classified as having prostate cancer,

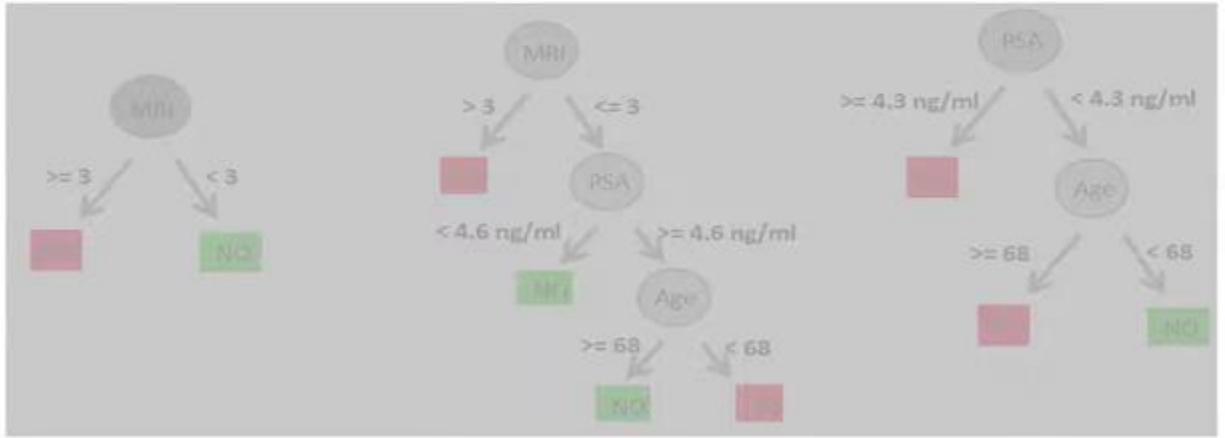
Estimate out-of-bag error

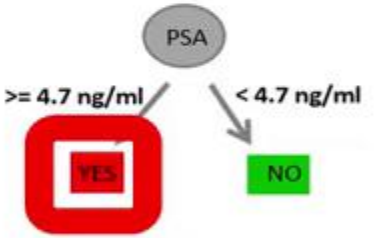
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Cancer

Healthy







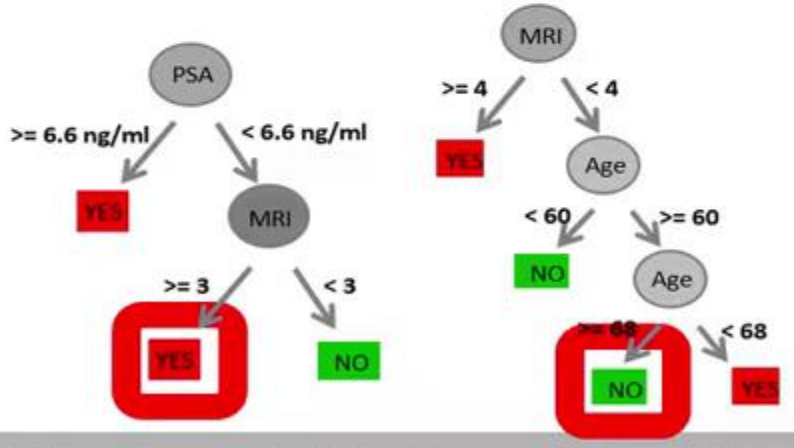
Since two out of the three trees predicted prostate cancer, the majority vote is for prostate cancer. The first case is therefore correctly predicted to have prostate cancer.

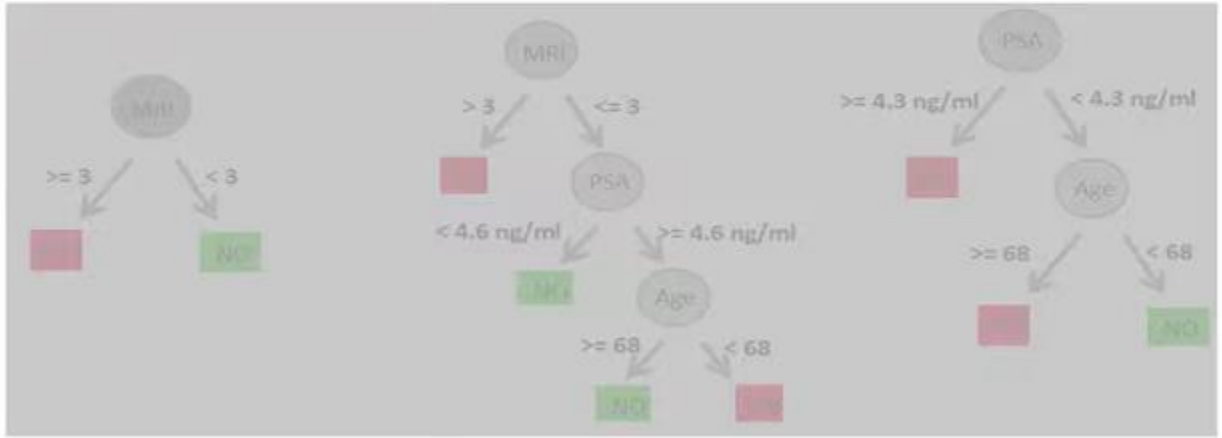
Estimate out-of-bag error

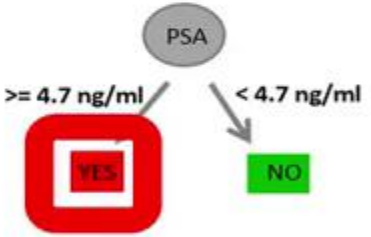
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2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

Cancer

Healthy







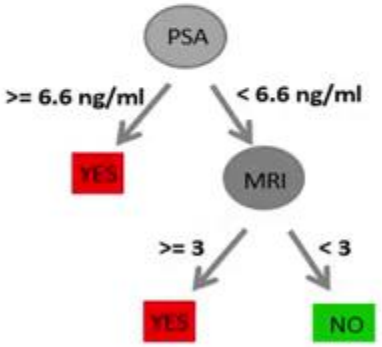
Since two out of the three trees predicted prostate cancer, the majority vote is for prostate cancer. The first case is therefore correctly predicted to have prostate cancer.

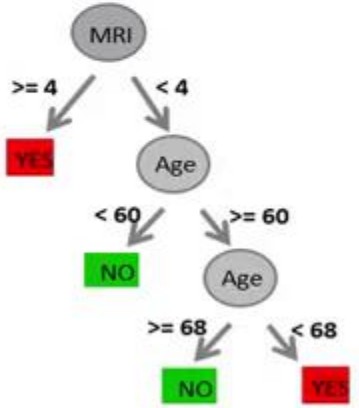
Estimate out-of-bag error

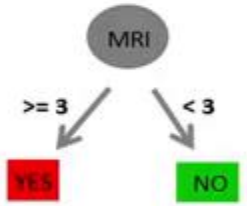
Person	Age	PSA	MRI
1	85	5.1	2
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

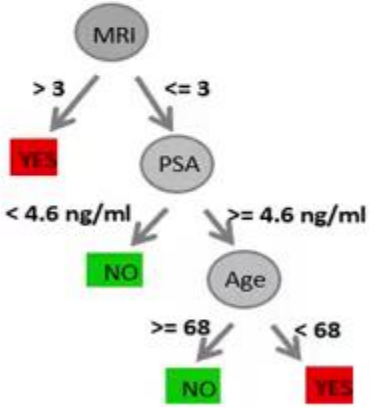
Cancer

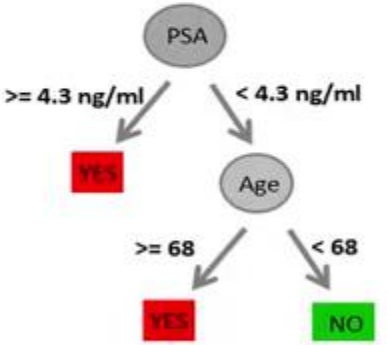
Healthy

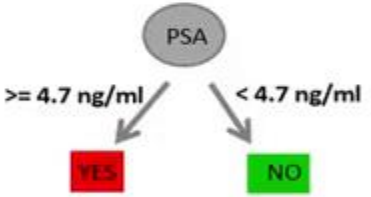












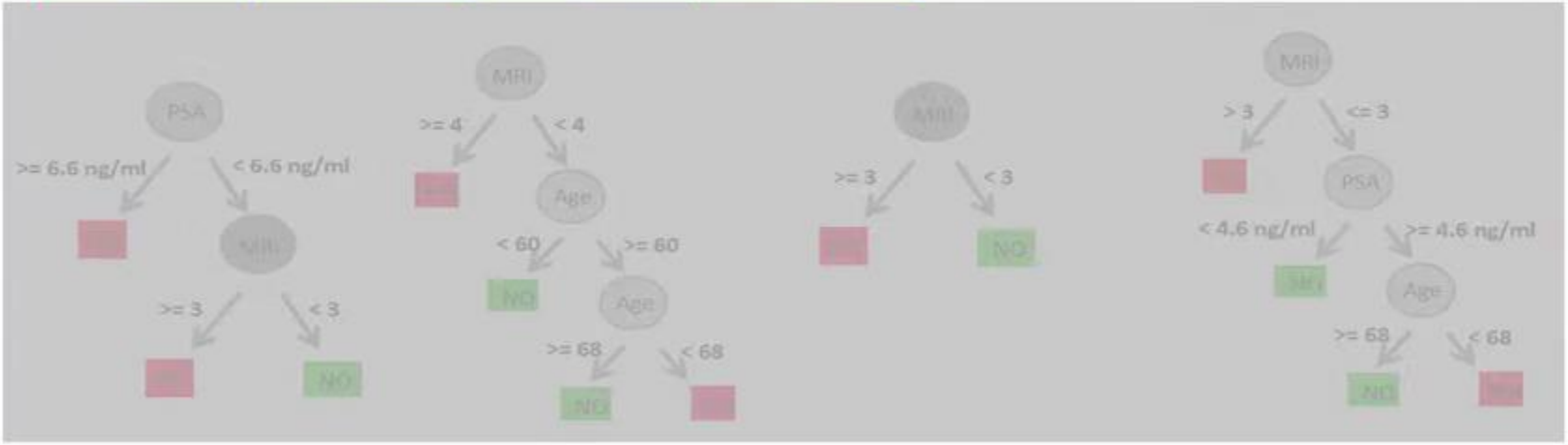
Then we do the same for the second case.

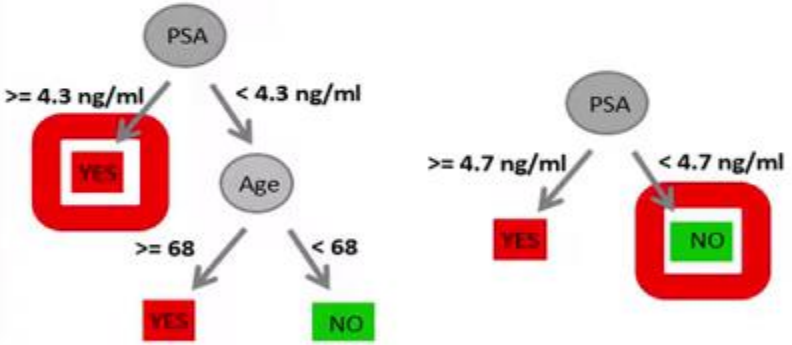
Estimate out-of-bag error

Person	Age	PSA	MRI
1	85	5.1	2
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

Cancer

Healthy





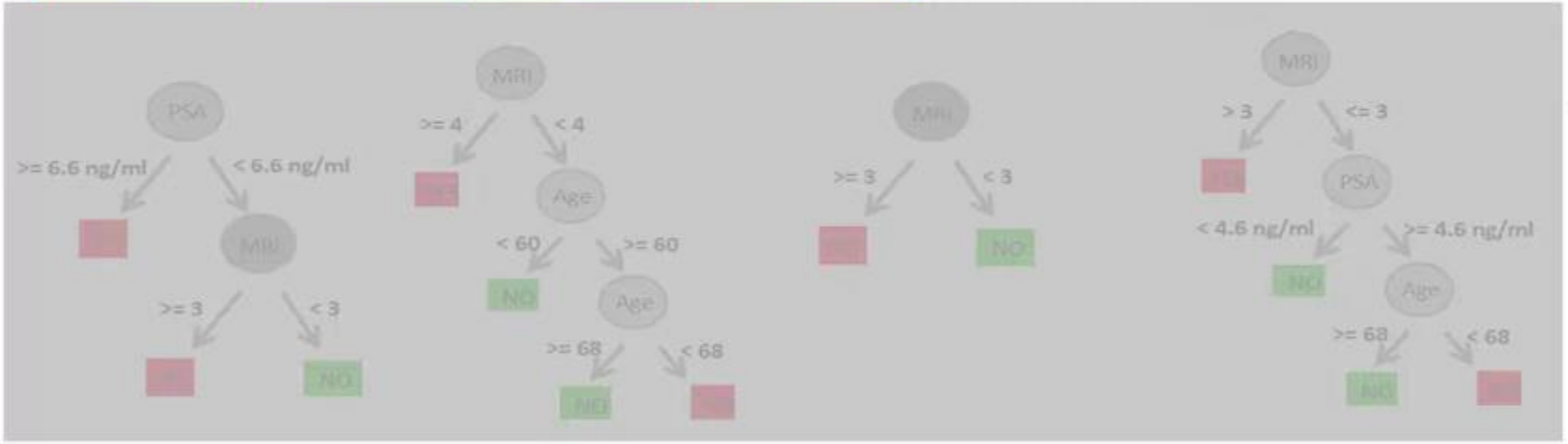
In this example, one tree predicted prostate cancer whereas the other tree predicted healthy. In this case, we have a tie. A tie can be broken at random or avoided by only allowing an odd number of trees.

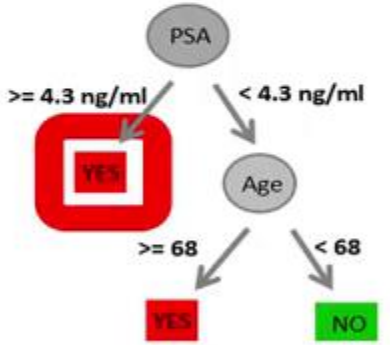
Estimate out-of-bag error

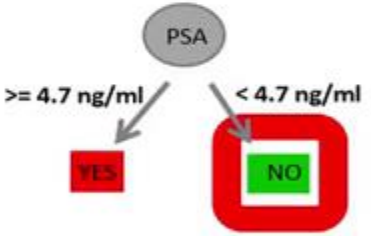
Person	Age	PSA	MRI
1	85	5.1	2
2	54	4.3	2
3	50	1.2	1
4	65	4.9	3
5	67	5.6	4
6	56	6.7	5
7	63	6.5	2
8	70	3.0	3
9	50	4.9	3

Cancer

Healthy







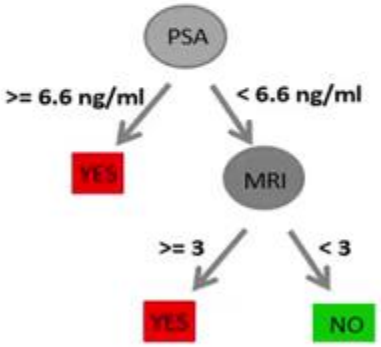
In this example, one tree predicted prostate cancer whereas the other tree predicted healthy. In this case, we have a tie. A tie can be broken at random or avoided by only allowing an odd number of trees.

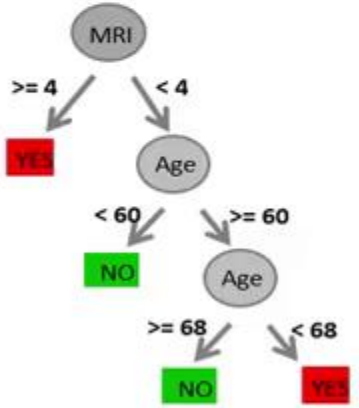
Estimate out-of-bag error

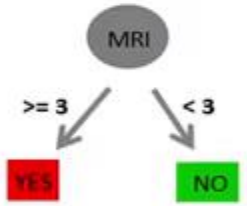
Person	Age	PSA	MRI	Predict
1	85	5.1	3	Cancer
2	54	4.3	2	Healthy
3	50	1.2	1	Cancer
4	65	4.9	3	Healthy
5	67	5.6	4	Cancer
6	56	6.7	5	Healthy
7	63	6.5	2	Healthy
8	70	3.0	3	Cancer
9	50	4.9	3	Healthy

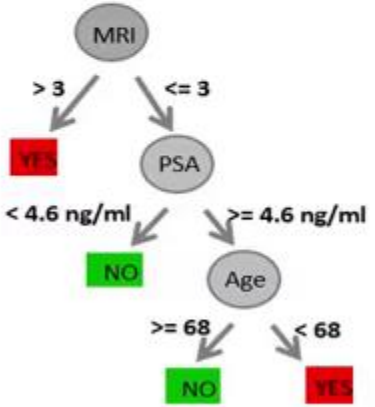
Cancer

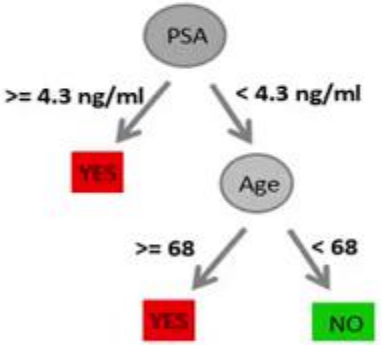
Healthy

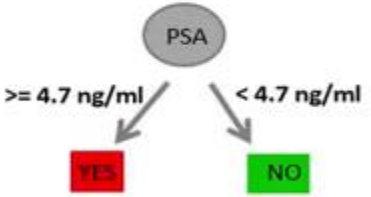












Let's say that the forest makes the following predictions.

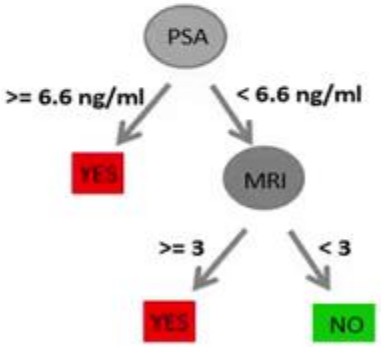
Estimate out-of-bag error

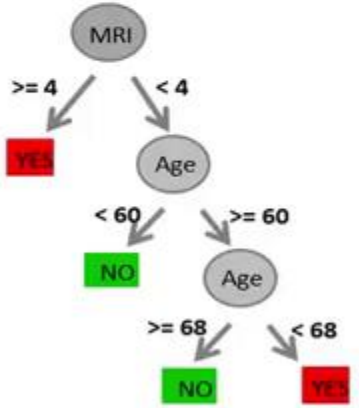
Person	Age	PSA	MRI	Predict
1	85	5.1	3	Cancer
2	54	4.3	2	Healthy
3	50	1.2	1	Cancer
4	65	4.9	3	Healthy
5	67	5.6	4	Cancer
6	56	6.7	5	Healthy
7	63	6.5	2	Healthy
8	70	3.0	3	Cancer
9	50	4.9	3	Healthy

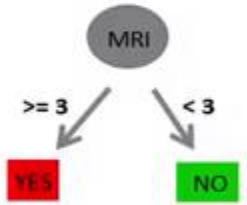
Cancer

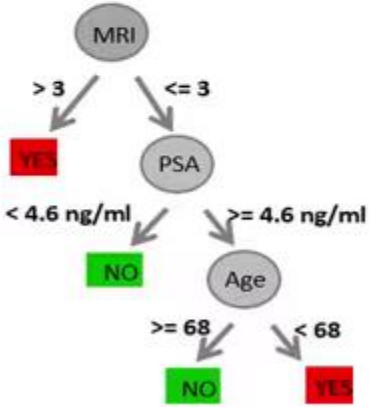
Healthy

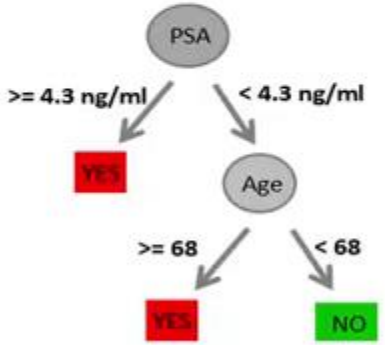
$$\frac{4}{9} = 44\%$$

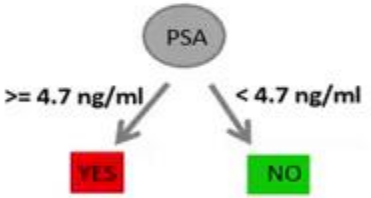










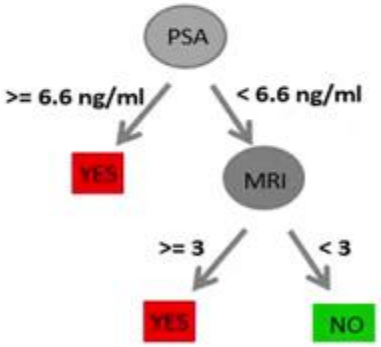


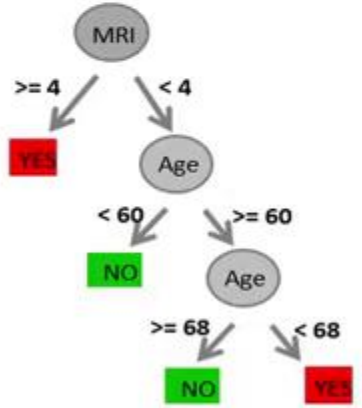
Since 4 out of the 9 cases were incorrectly predicted, this means that the out-of-bag error is equal to about 44%.

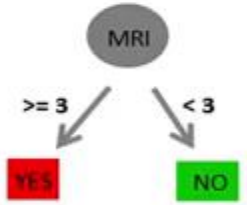
Estimate out-of-bag error

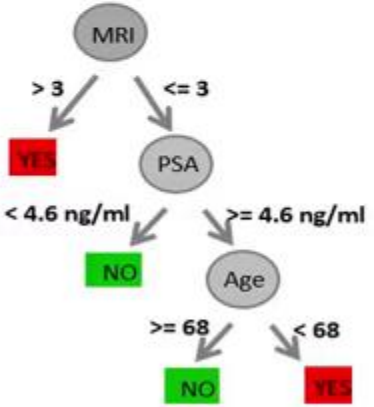
Cancer
Healthy

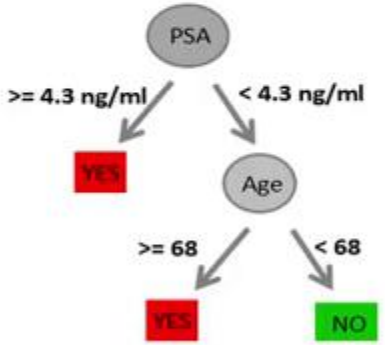
$$\frac{4}{9} = 44\%$$

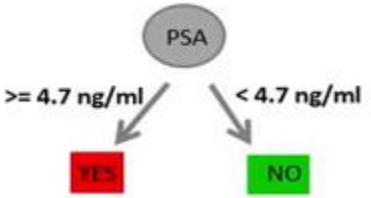










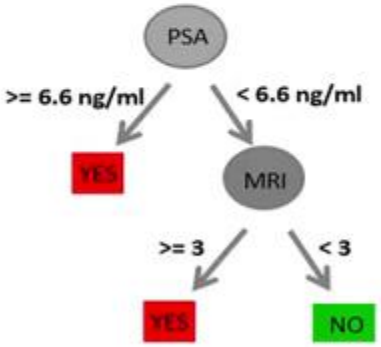


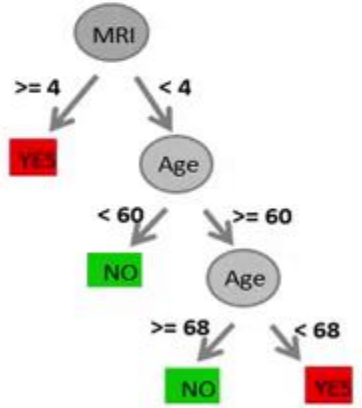
Our forest is therefore not useful in predicting prostate cancer based on this example data. This is mainly due to that our sample size is too small for classification.

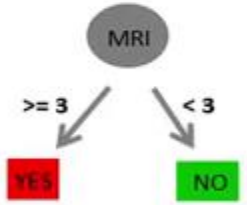
Estimate out-of-bag error

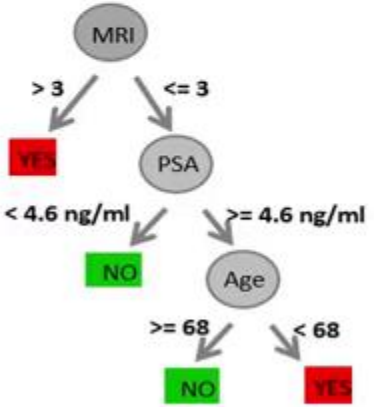
Cancer
Healthy

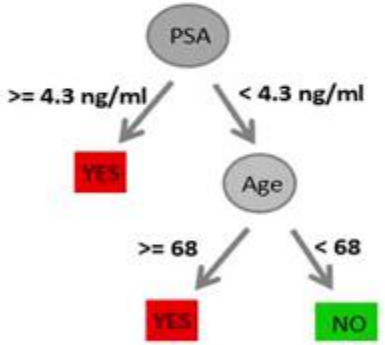
$$\frac{4}{9} = 44\%$$

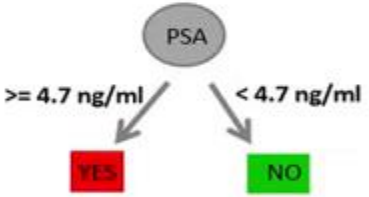










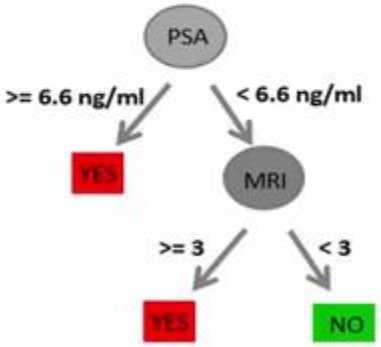


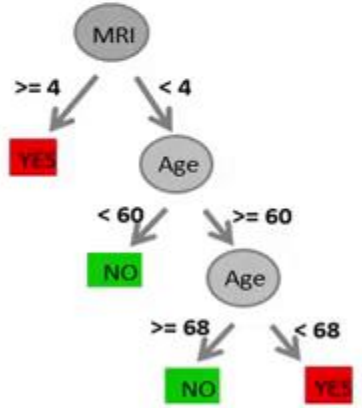
What is nice about this method is that the out-of-bag data works as a test data set, which means that we both train and validate our method during the procedure.

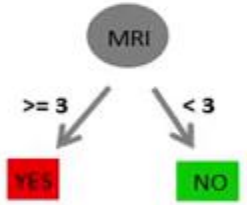
Estimate out-of-bag error

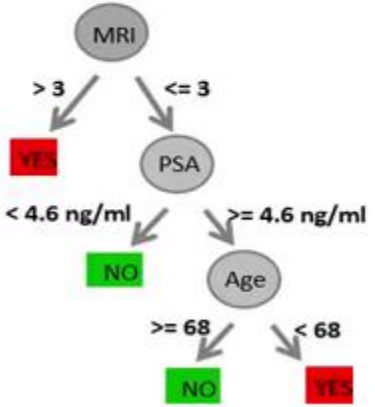
Cancer
Healthy

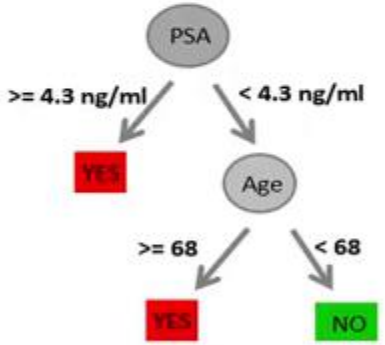
$$\frac{4}{9} = 44\%$$

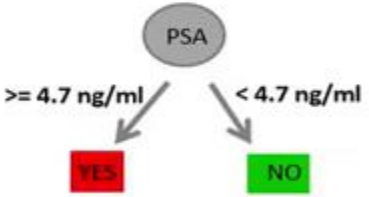






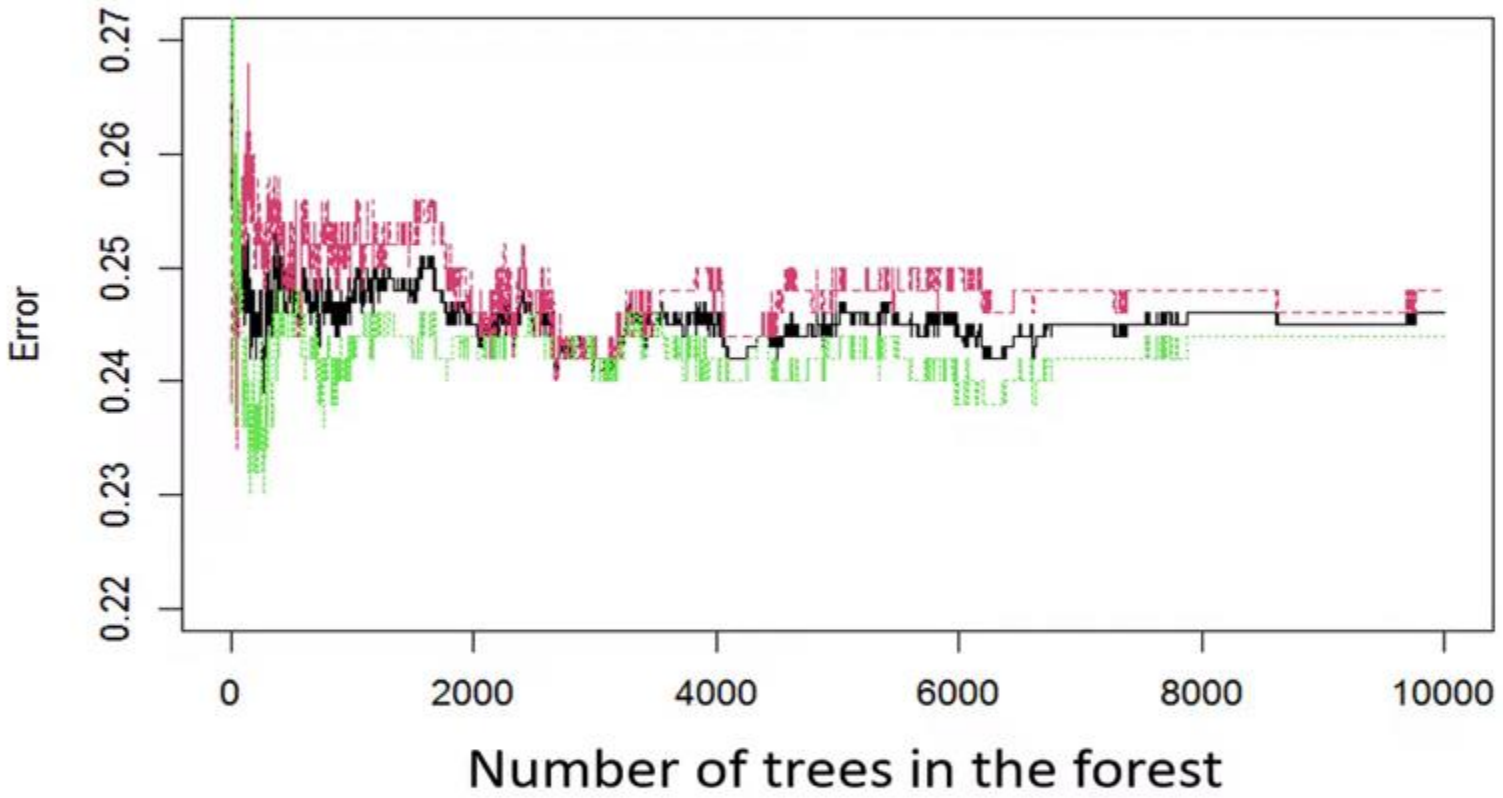






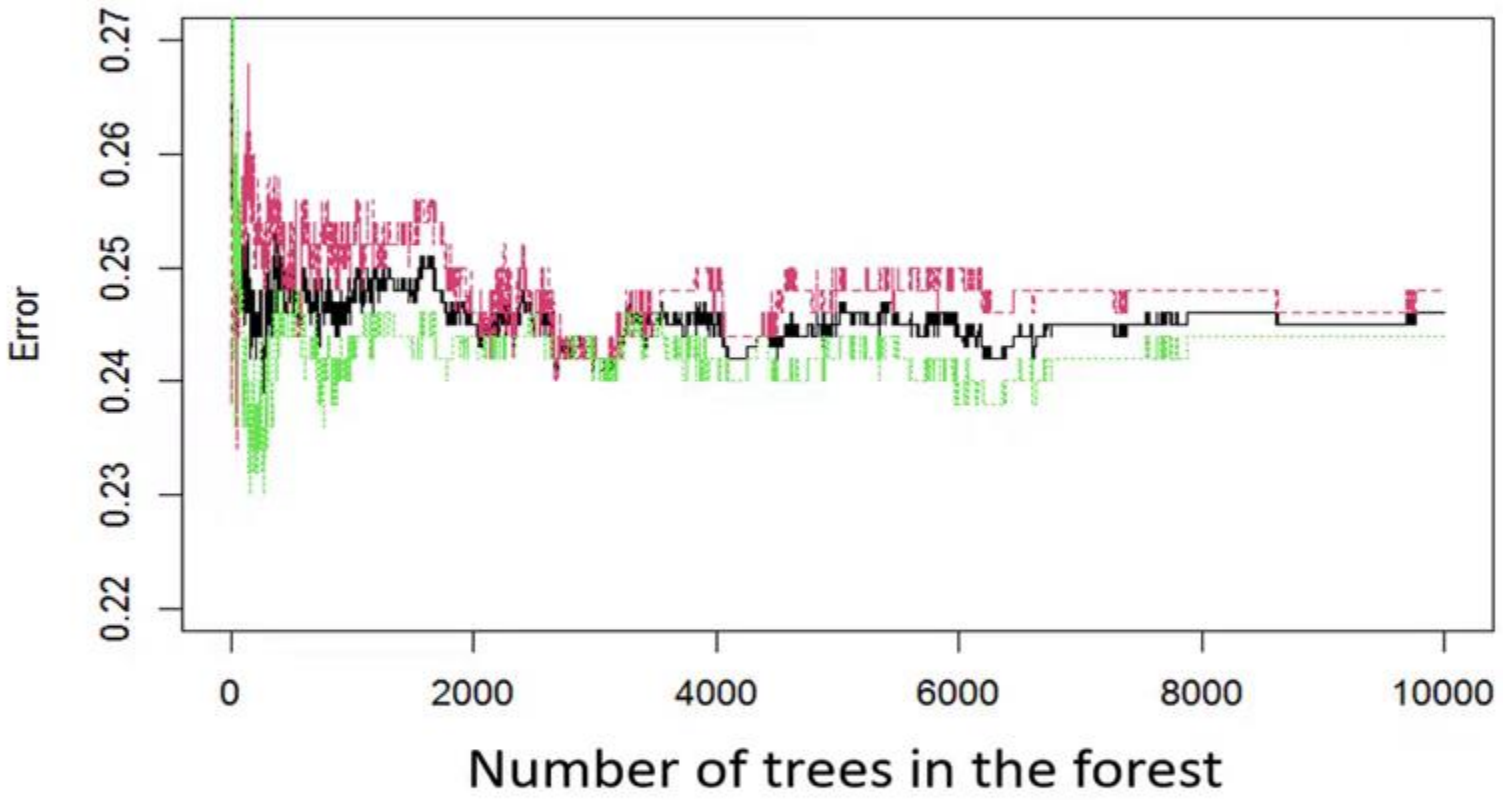
What is nice about this method is that the out-of-bag data works as a test data set, which means that we both train and validate our method during the procedure.

Number of trees



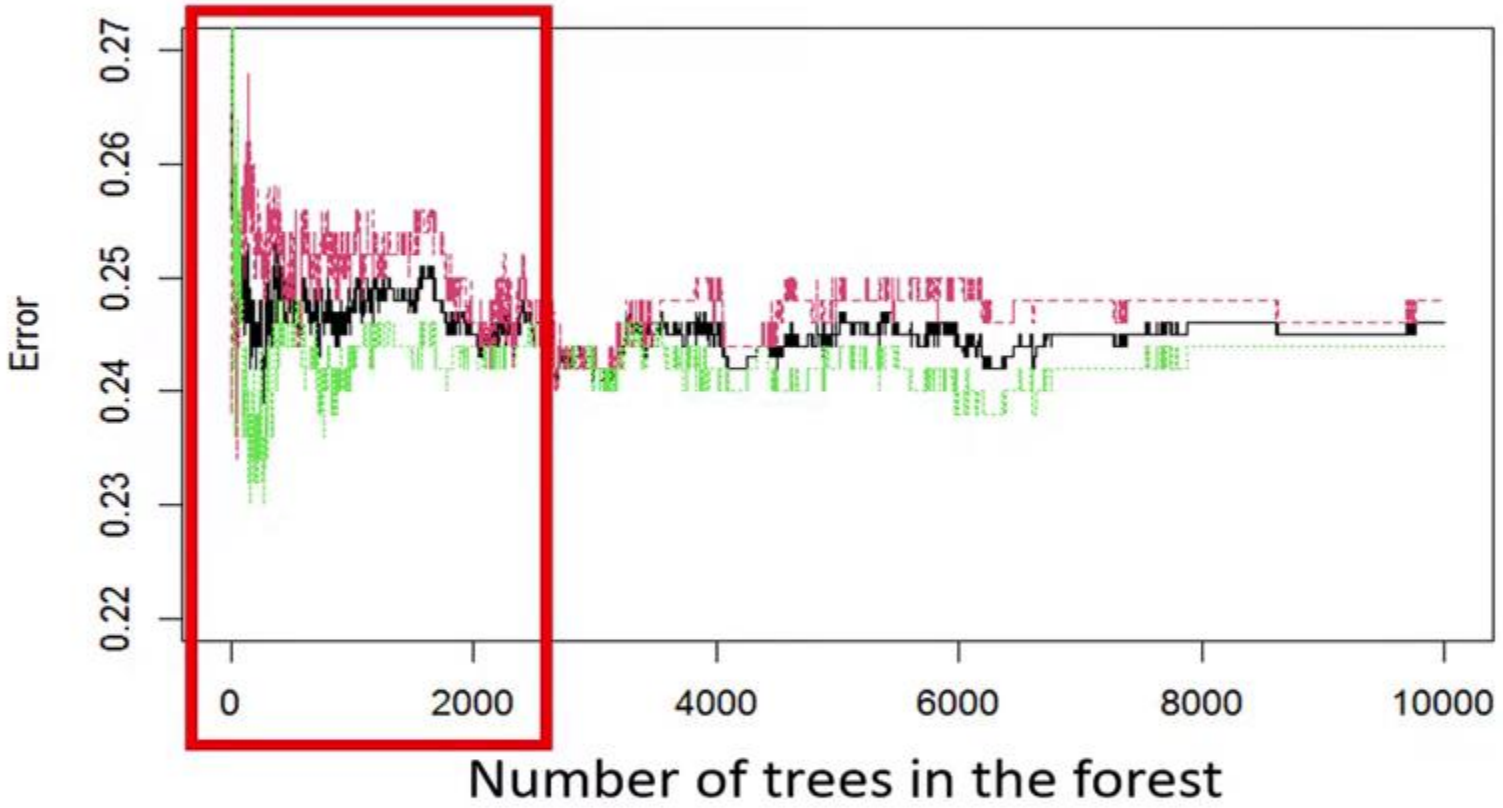
To know how many trees we should have in our forest, we can study how the out-of-bag error changes as the number of trees in the forest increase. The black line in this plot shows the average error of the two groups for a larger data set.

Number of trees



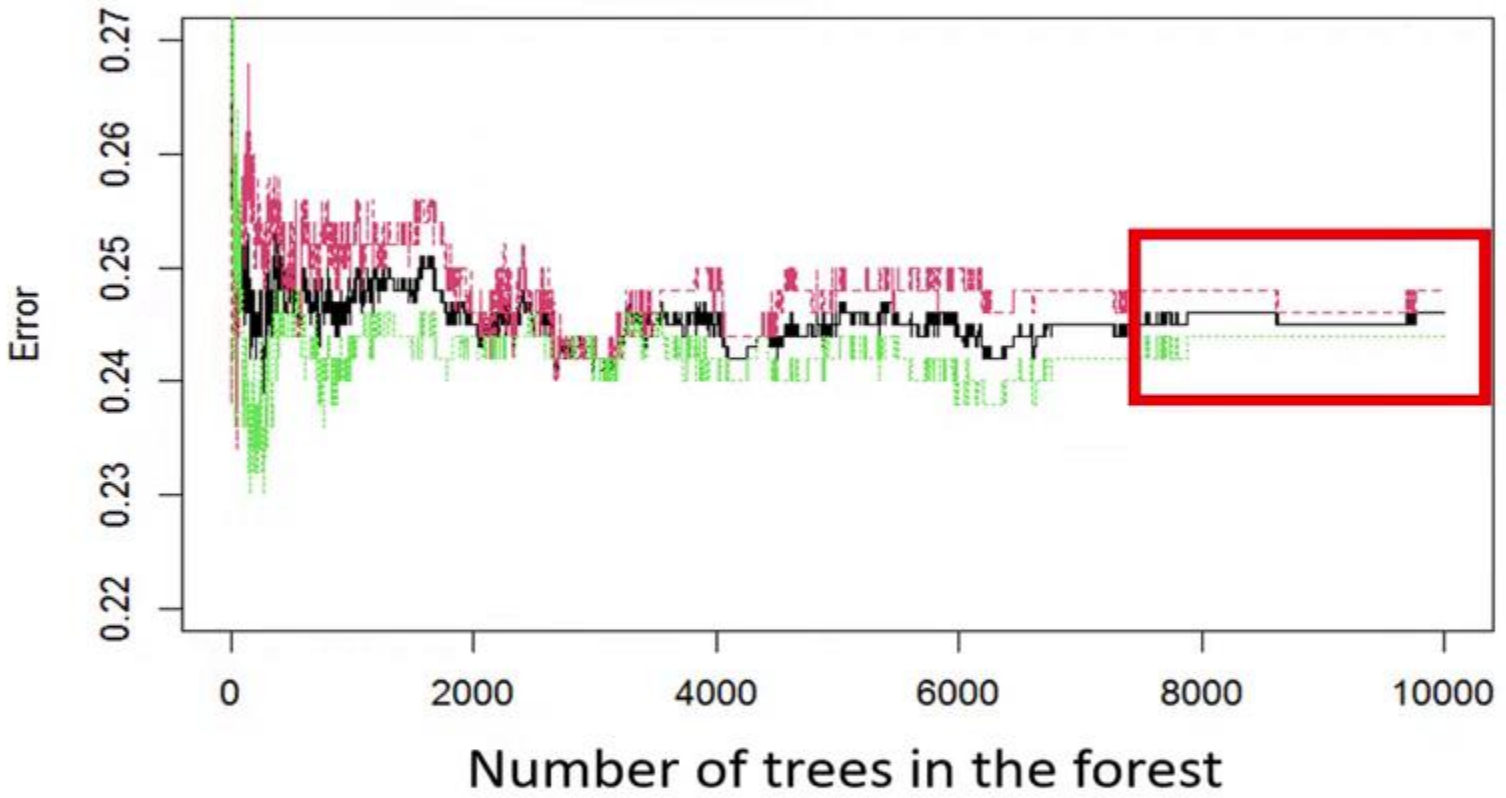
To know how many trees we should have in our forest, we can study how the out-of-bag error changes as the number of trees in the forest increase. The black line in this plot shows the average error of the two groups for a larger data set.

Number of trees



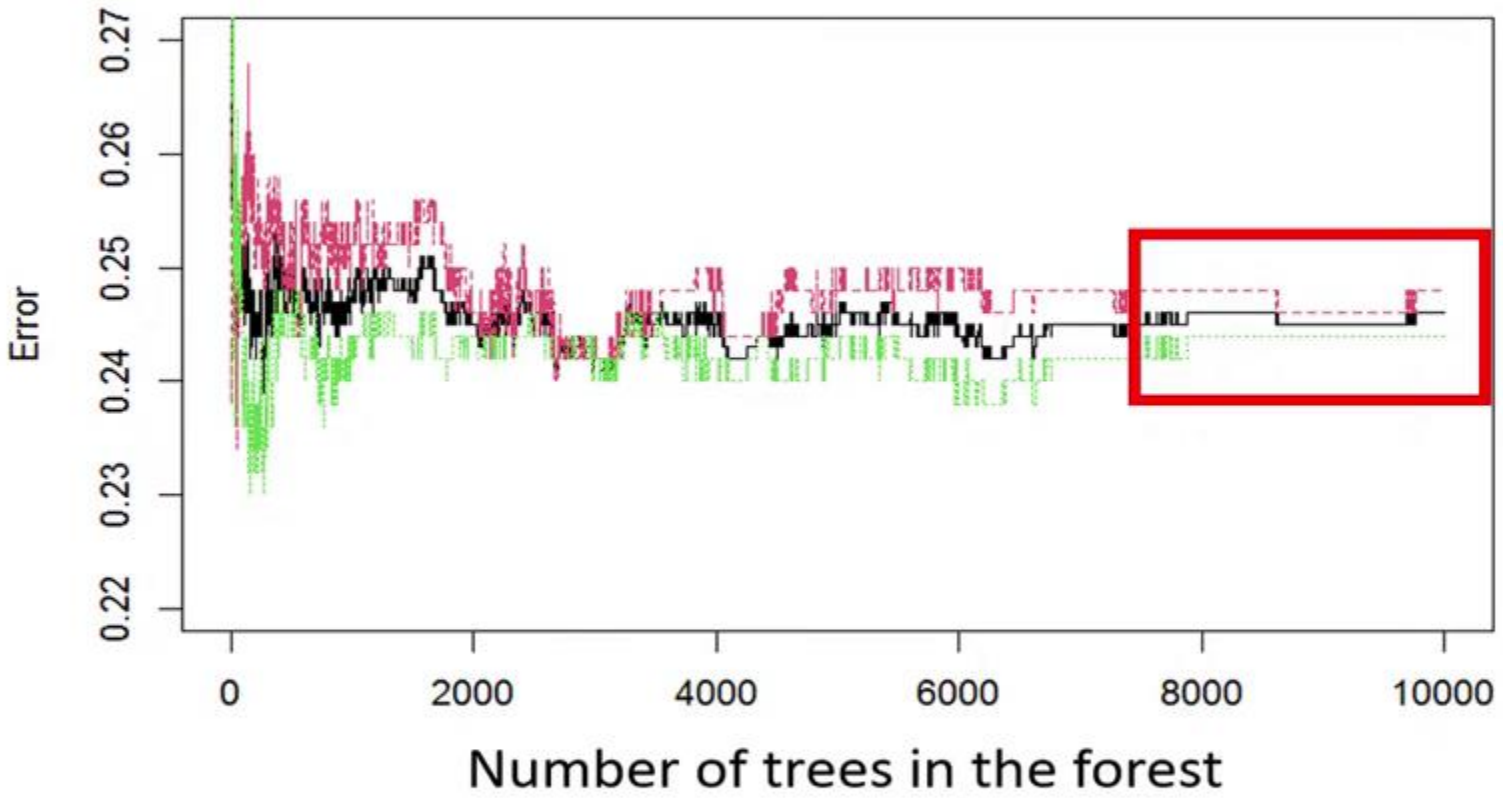
We see that the error varies a lot when we have relatively few trees in the forest,

Number of trees



whereas the error seems to stabilize when the forest consists of more than about 8000 trees. This shows that we should select at least 8000 trees for our forest, because we prefer a forest with a small variation.

Number of trees

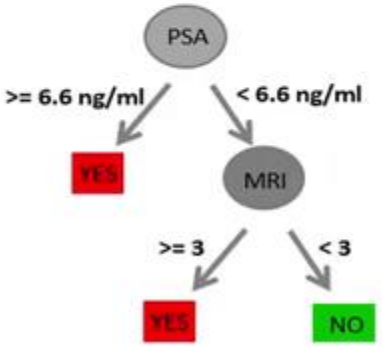


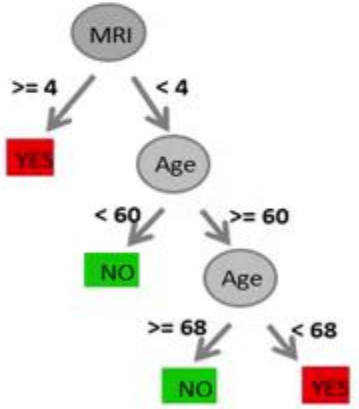
whereas the error seems to stabilize when the forest consists of more than about 8000 trees. This shows that we should select at least 8000 trees for our forest, because we prefer a forest with a small variation.

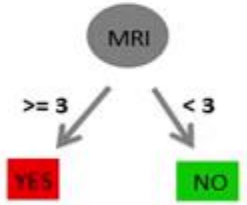
Prediction

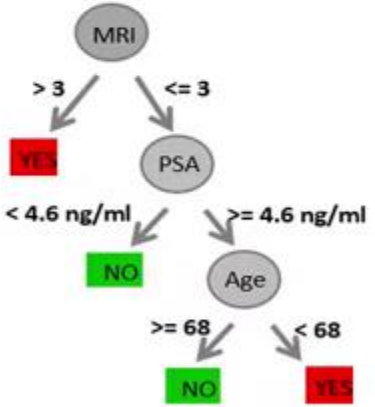
Age	PSA	MRI
69	6.1	3

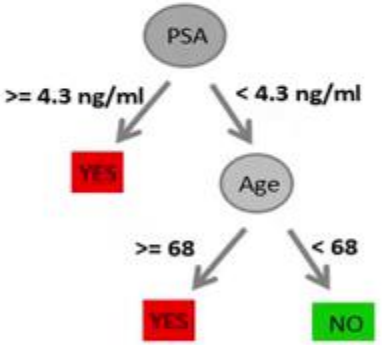
Cancer
Healthy

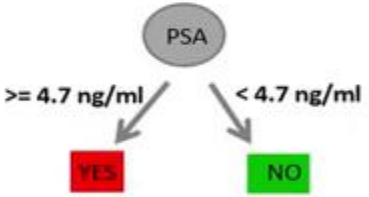












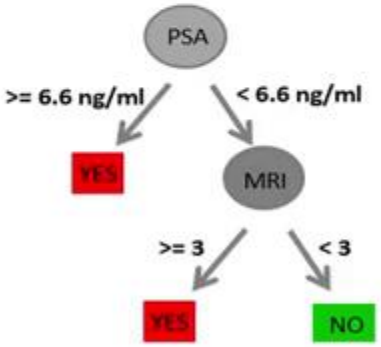
Let's have a look at how the random forest would predict a new case. In this example, we therefore do not know if the person has prostate cancer or not. We will therefore let the forest predict this for us.

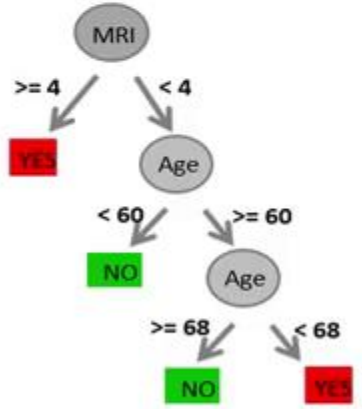
Prediction

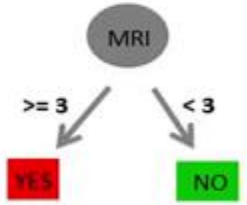
Age	PSA	MRI
69	6.1	3

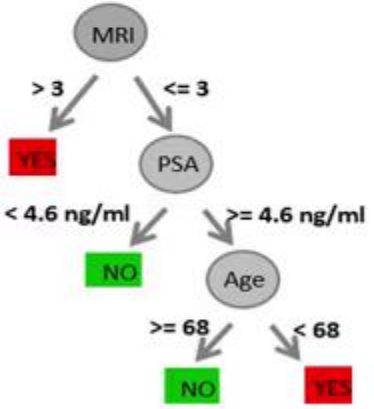
Cancer

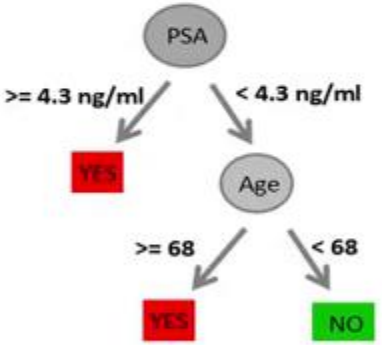
Healthy

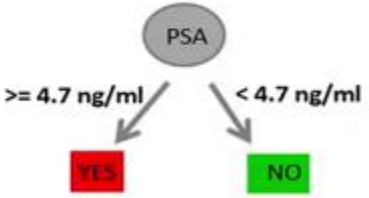












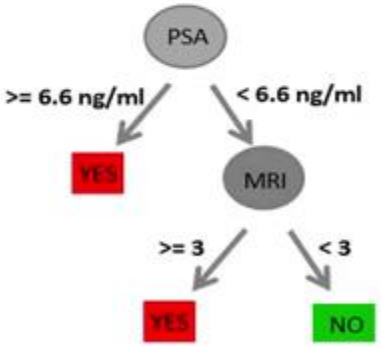
Let's have a look at how the random forest would predict a new case. In this example, we therefore do not know if the person has prostate cancer or not. We will therefore let the forest predict this for us.

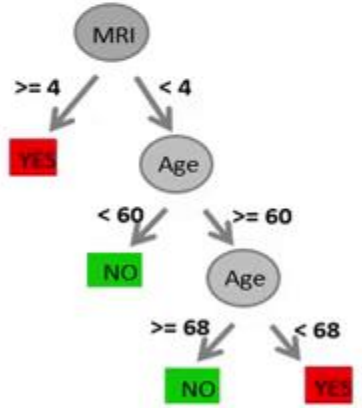
Prediction

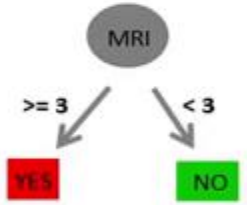
Age	PSA	MRI
69	6.1	3

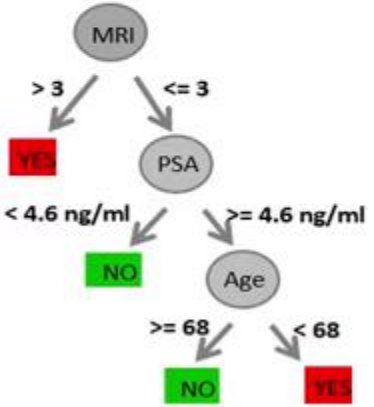
Cancer

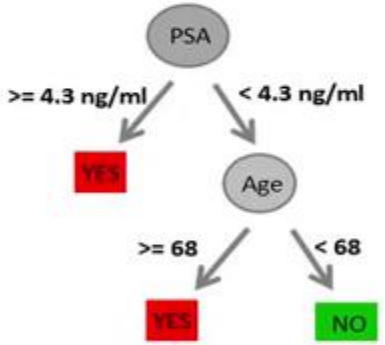
Healthy

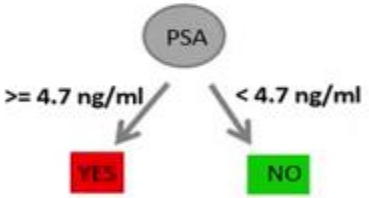












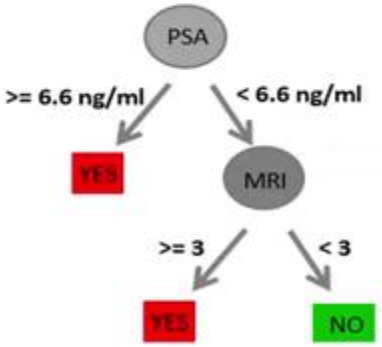
The person has an age of 69, a PSA level of 6.1, and an MRI score of three.

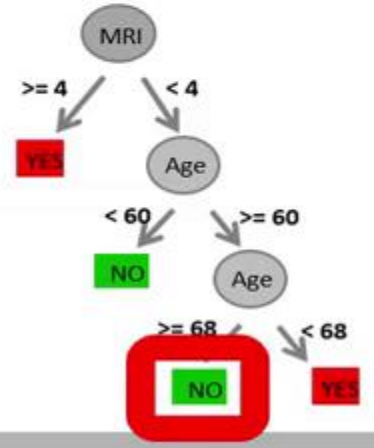
Prediction

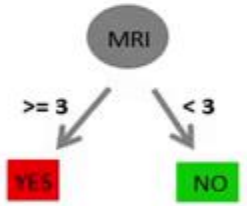
Age	PSA	MRI
69	6.1	3

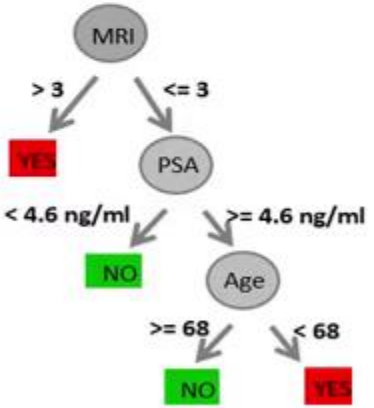
Cancer

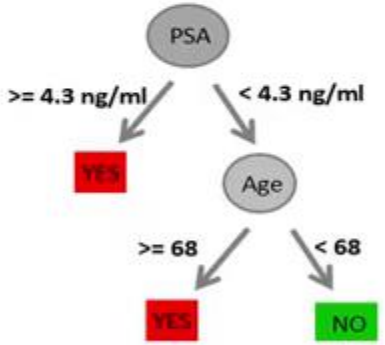
Healthy

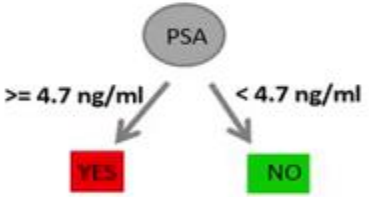












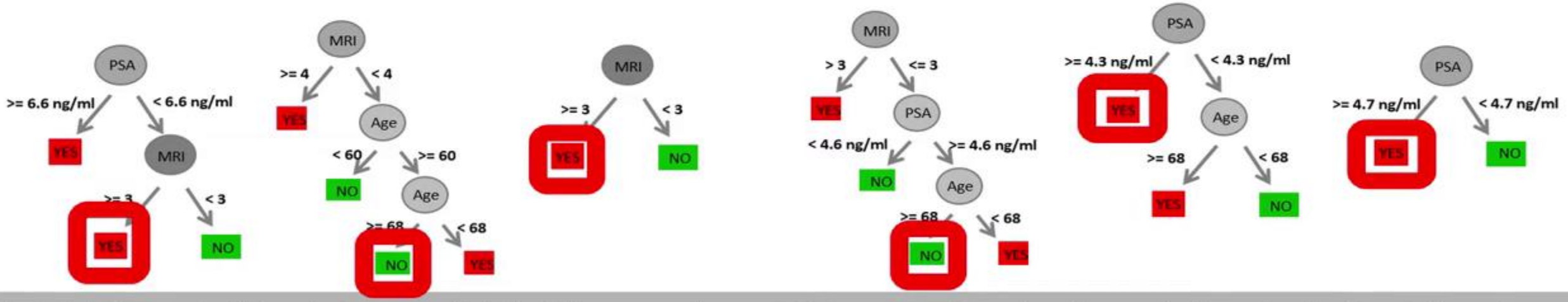
whereas the second tree will predict healthy, and so forth.

Prediction

Age	PSA	MRI
69	6.1	3

Cancer

Healthy



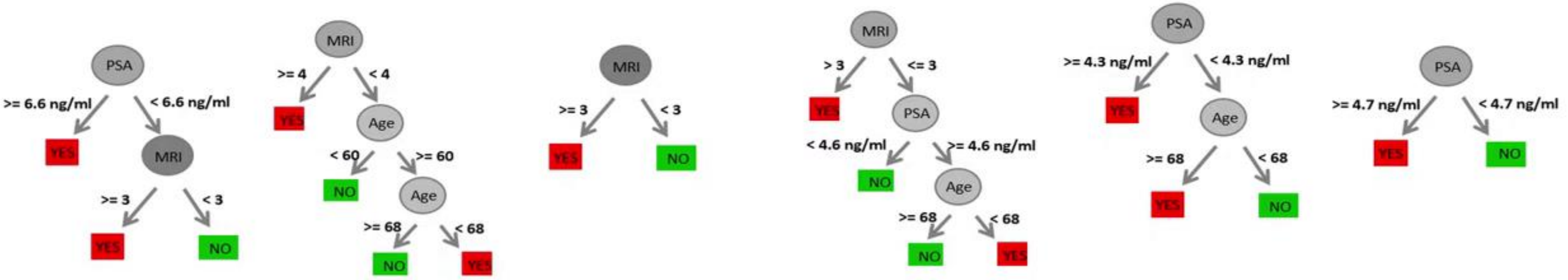
In total, four out of the six trees predict that the person has prostate cancer. Based on the majority vote, the forest therefore predicts that the person has prostate cancer.

Variable importance

Variable importance

Cancer

Healthy



One nice feature of random forest is that it can give us a measure of how well a variable can discriminate between the two classes.

Variable importance

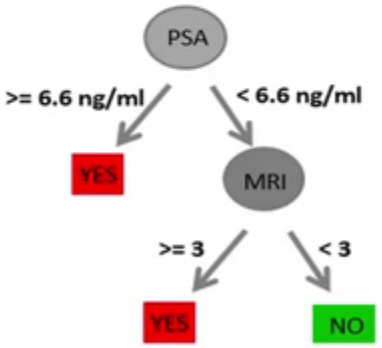
Mean decrease in Gini index for MRI = 0.9

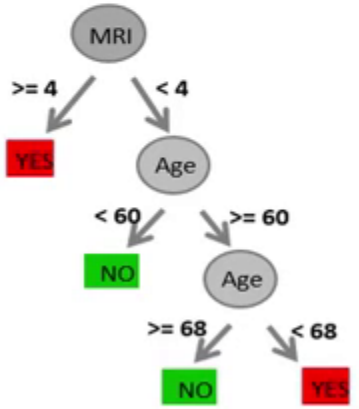
Mean decrease in Gini index for PSA = 1.3

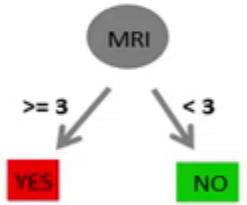
Mean decrease in Gini index for Age = 1.6

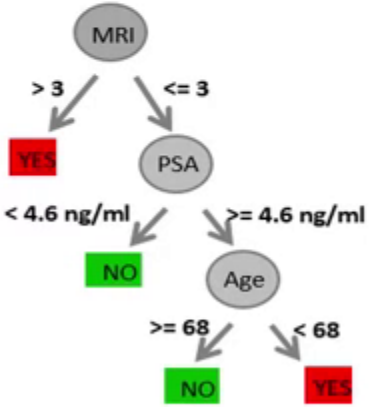
Cancer

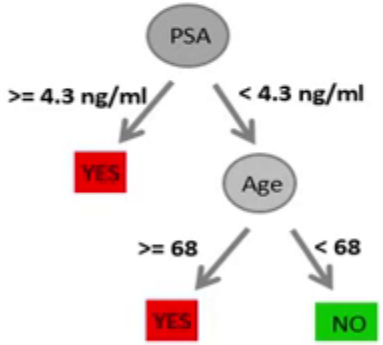
Healthy

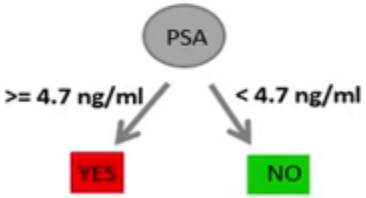












One such measure is the mean decrease in the Gini index.

Variable importance

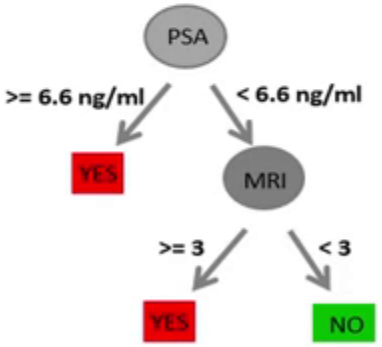
Mean decrease in Gini index for MRI = 0.9

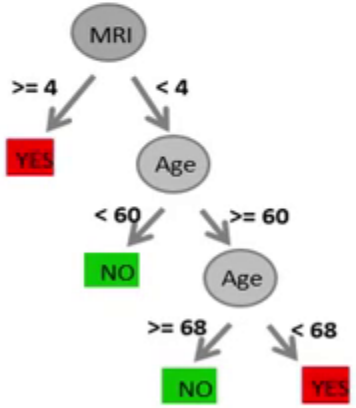
Mean decrease in Gini index for PSA = 1.3

Mean decrease in Gini index for Age = 1.6

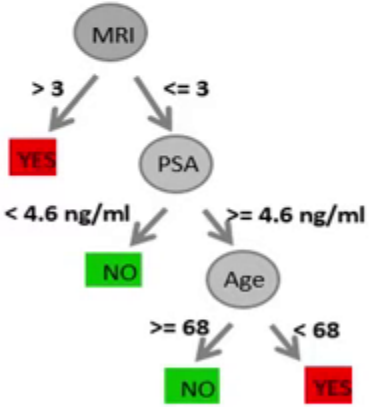
Cancer

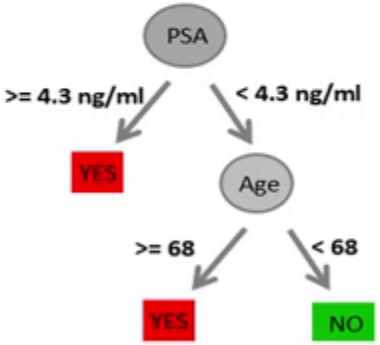
Healthy

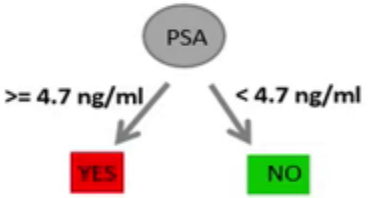












For example, the mean decrease in the Gini index for the variable MRI is equal to 0.9,

Variable importance

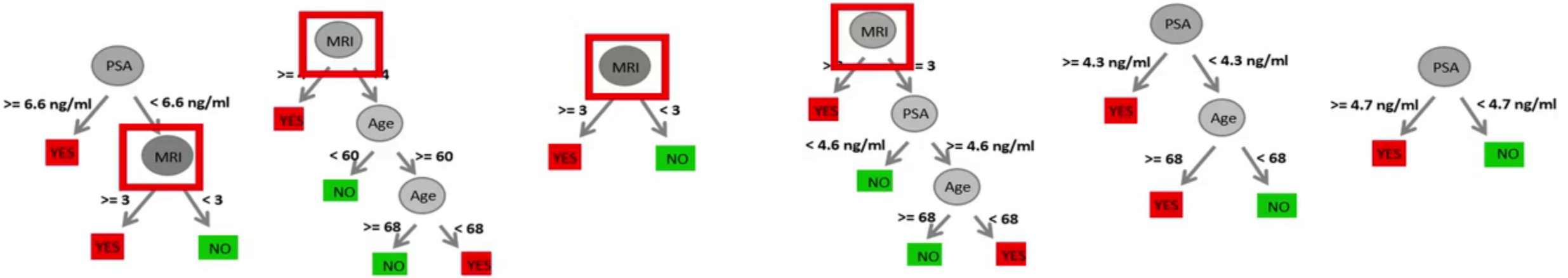
Mean decrease in Gini index for MRI = 0.9

Mean decrease in Gini index for PSA = 1.3

Mean decrease in Gini index for Age = 1.6

Cancer

Healthy



which is based on the Gini index of the following four splits, if our forest consists of only these six trees.

Variable importance

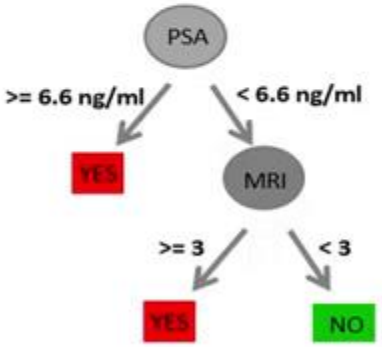
Mean decrease in Gini index for MRI = 0.9

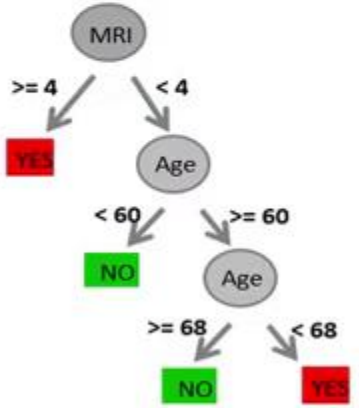
Mean decrease in Gini index for PSA = 1.3

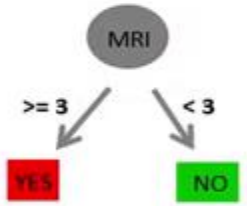
Mean decrease in Gini index for Age = 1.6

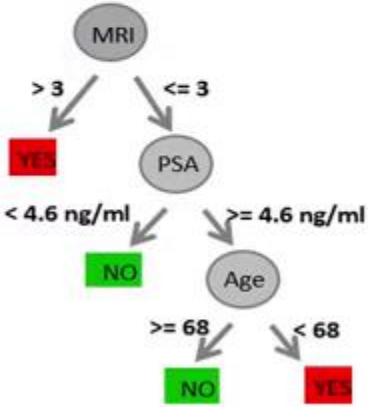
Cancer

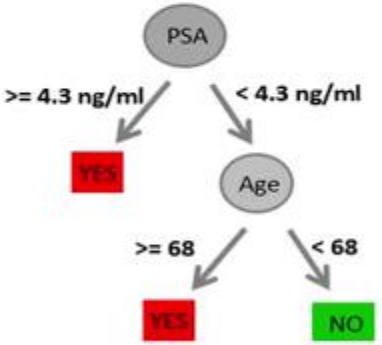
Healthy

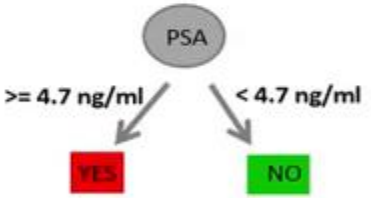












A good variable should have a relatively high value because it is then good at decreasing the Gini index, and therefore good at creating as pure groups as possible.

Variable importance

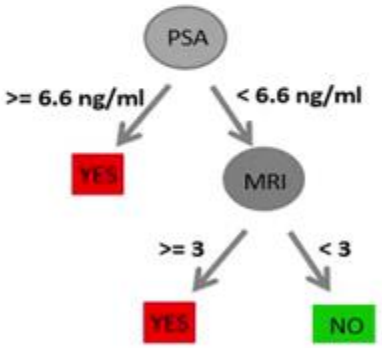
Mean decrease in Gini index for MRI = 0.9

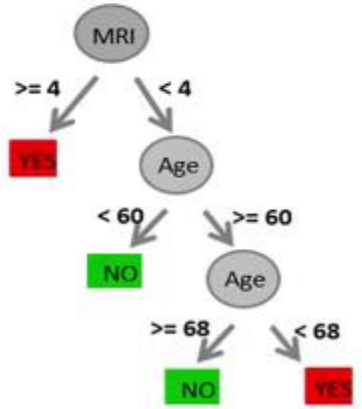
Mean decrease in Gini index for PSA = 1.3

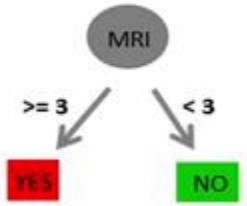
Mean decrease in Gini index for Age = 1.6

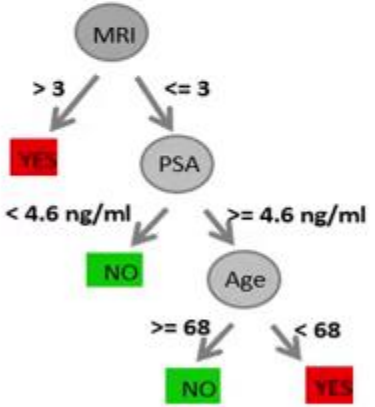
Cancer

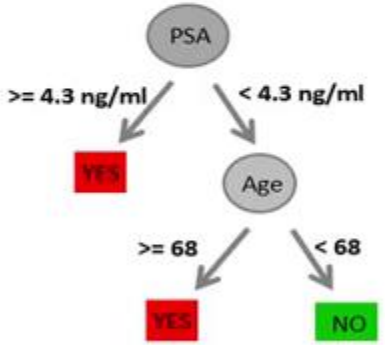
Healthy

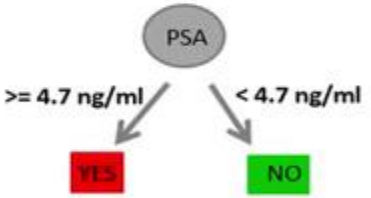












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Variable importance

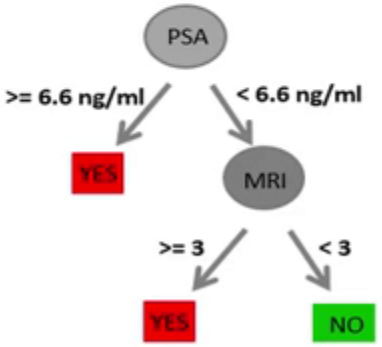
Mean decrease in Gini index for MRI = 0.9

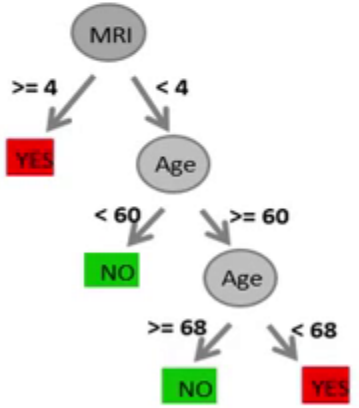
Mean decrease in Gini index for PSA = 1.3

Mean decrease in Gini index for Age = 1.6

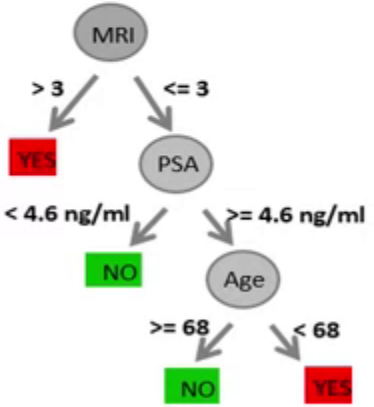
Cancer

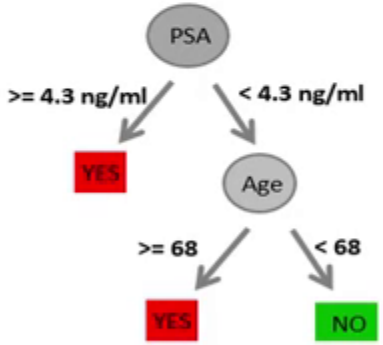
Healthy

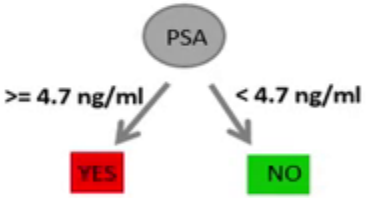












In this case, we see that Age is the best variable to discriminate the cancer cases from the healthy controls.

Variable importance

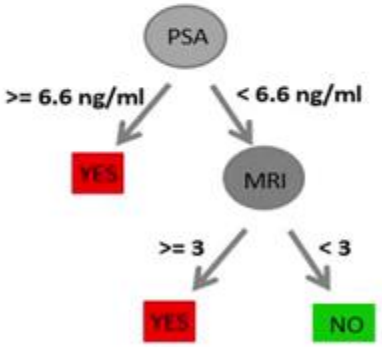
Mean decrease in Gini index for MRI = 0.9

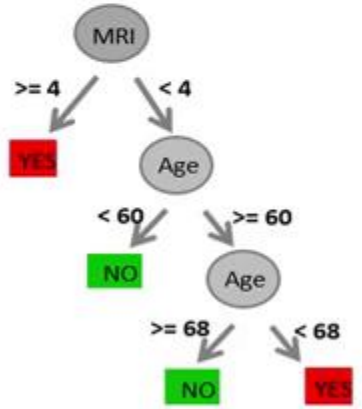
Mean decrease in Gini index for PSA = 1.3

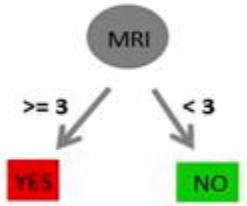
Mean decrease in Gini index for Age = 1.6

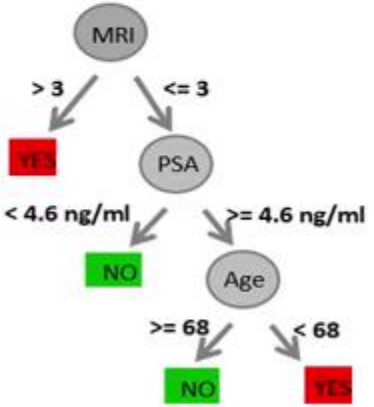
Cancer

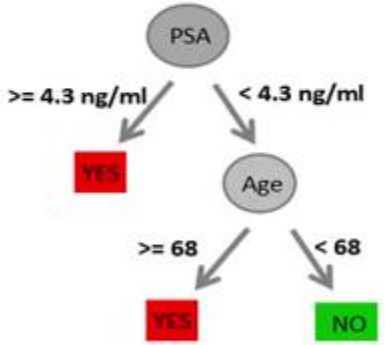
Healthy

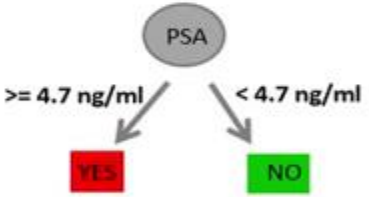












Calculating the variable importance is very useful when we have hundreds of variables and would like to extract a few that are good in separating the groups.

Variable importance

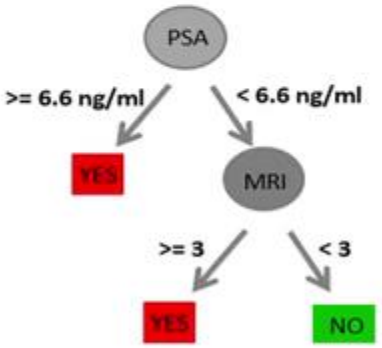
Mean decrease in Gini index for MRI = 0.9

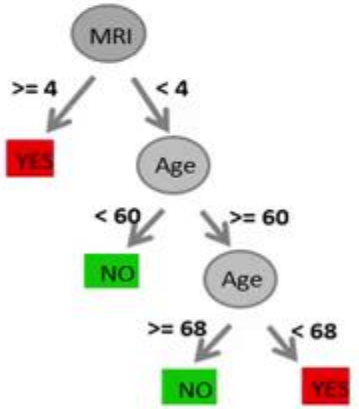
Mean decrease in Gini index for PSA = 1.3

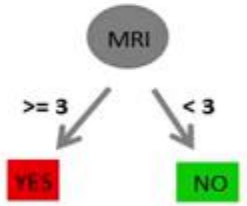
Mean decrease in Gini index for Age = 1.6

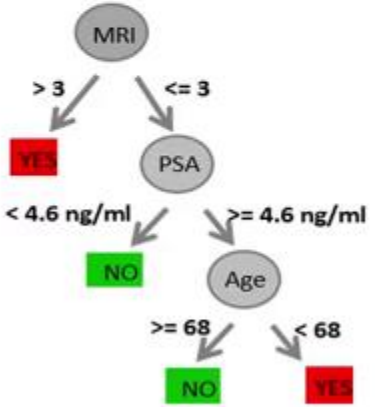
Cancer

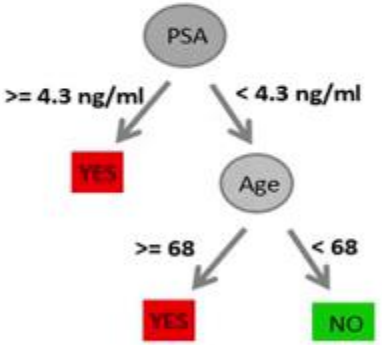
Healthy

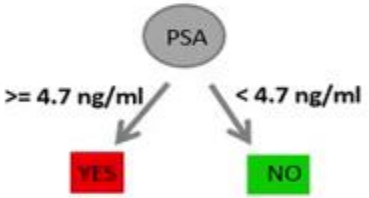












Calculating the variable importance is very useful when we have hundreds of variables and would like to extract a few that are good in separating the groups.

Pros and cons

Random forest vs decision tree

Pros and cons

Random forest vs decision tree

+ No need to run cross-validation or use a separate test data set

One advantage is that there is no need to run cross-validation or use a separate test data set in a random forest because it internally estimates the error rate by the out-of-bag error.

Pros and cons

Random forest vs decision tree

+ No need to run cross-validation or use a separate test data set

+ Higher accuracy

A random forest generally results in higher accuracy, or less error, than decision trees.

Pros and cons

Random forest vs decision tree

+ No need to run cross-validation or use a separate test data set

+ Higher accuracy

+ Does not overfit -> no need for pruning

In comparison to decision trees, which are highly likely to overfit, a random forest avoids this problem by using many trees that are randomly generated.

Pros and cons

Random forest vs decision tree

+ No need to run cross-validation or use a separate test data set

+ Higher accuracy

+ Does not overfit -> no need for pruning

+ Variable (or feature) importance

A random forest also generates a measure of variable importance for classification, that can be used for feature selection,

Pros and cons

Random forest vs decision tree

+ No need to run cross-validation or use a separate test data set

+ Higher accuracy

+ Does not overfit -> no need for pruning

+ Variable (or feature) importance

+ More robust (less variance)

and is more robust, or less sensitive, to the exact values in the training data.

Pros and cons

Random forest vs decision tree

+ No need to run cross-validation or use a separate test data set

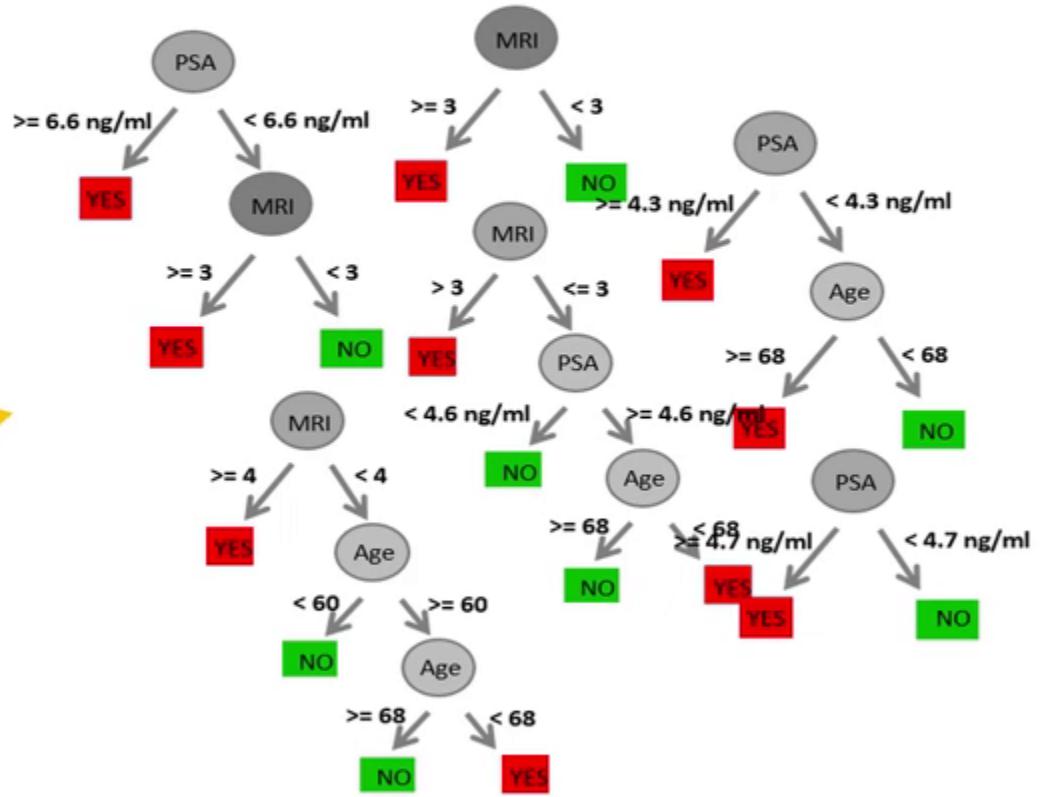
+ Higher accuracy

+ Does not overfit -> no need for pruning

+ Variable (or feature) importance

+ More robust (less variance)

- Harder to interpret



On the downside, it is much harder to understand how a forest predicts the class of a new case,

Pros and cons

Random forest vs decision tree

+ No need to run cross-validation or use a separate test data set

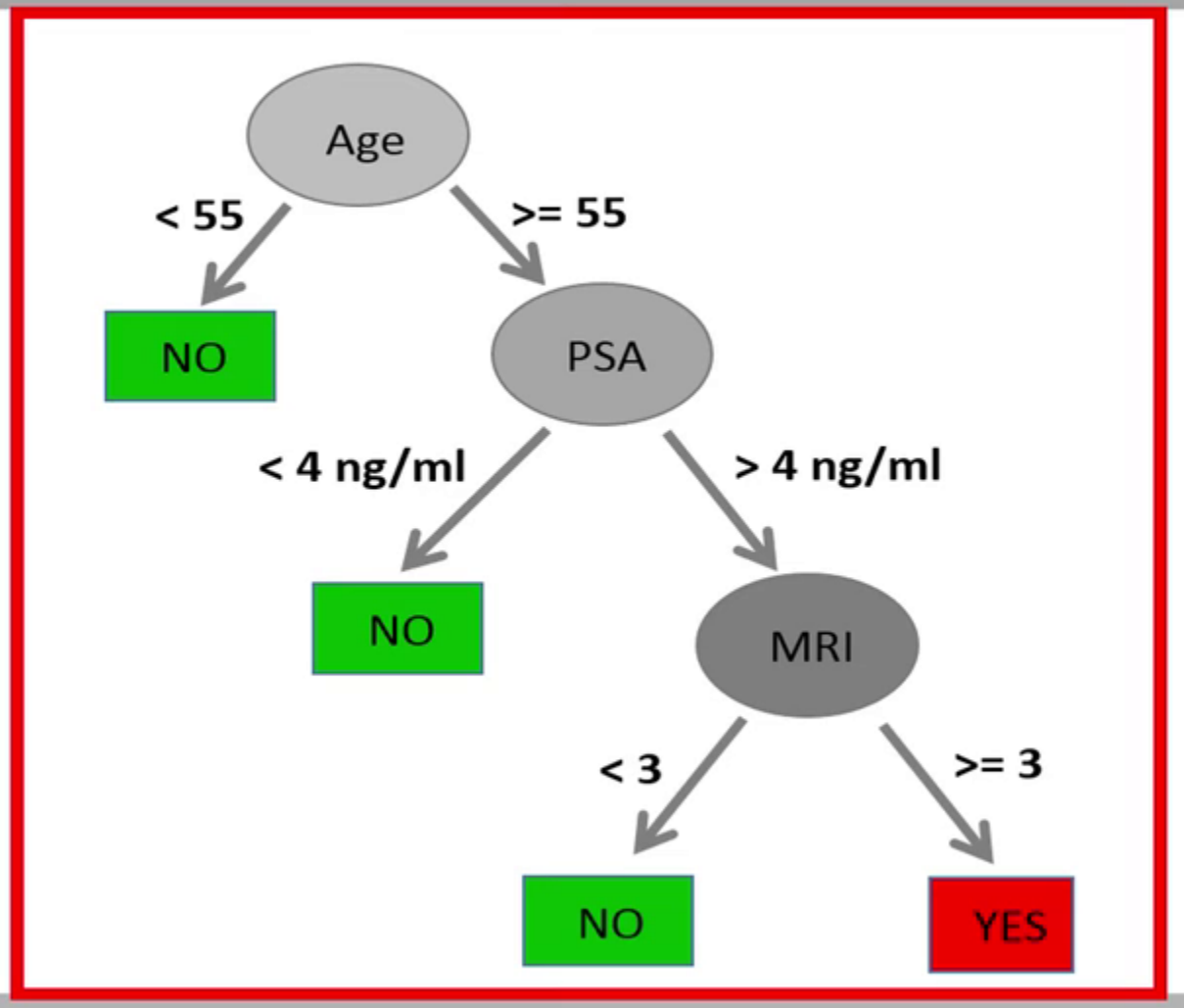
+ Higher accuracy

+ Does not overfit -> no need for pruning

+ Variable (or feature) importance

+ More robust (less variance)

- Harder to interpret



compared to a single decision tree.

Pros and cons

Random forest vs decision tree

+ No need to run cross-validation or use a separate test data set

+ Higher accuracy

+ Does not overfit -> no need for pruning

+ Variable (or feature) importance

+ More robust (less variance)

- Harder to interpret

- Computationally expensive

Also, since a random forest computes hundreds or thousands of trees, it is much more computationally expensive in comparison to a decision tree.

Pros and cons

Note that both decision trees and random forests work just fine with variables (features) on different scales, such as numerical and categorical variables, in comparison to most other classification methods. Also, there is usually no need to first normalize the data.

END

